



Implementation of the Fama-French factors

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List of Abbreviations

ADF	Augmented Dickey-Fuller
AMEX	American Stock Exchange
BE/ME	Book Equity to Market Equity
BTC	Bitcoin
CAPM	Capital Asset Pricing Model
C-CAPM	Conventional Capital Asset Pricing Model
F&F	Fama and French
FGLS	Feasible Generalized Least Squares
GRS	Gibbons-Ross-Shanken
HML	High Minus Low
IMI	Institutional Minus Individual
IRS	Internal Revenue Service
ISMS	Istanbul Stock Market Sustainability
JB	Jarque-Bera
Lbitcoin return	Logarithm of Bitcoin return
LHML	Logarithm of High Minus Low
Lmarket risk premium	Logarithm of market risk premium
LSMB	Logarithm of Small Minus Big
Lvwret_risk_premium	Logarithm of value-weighted average return minus risk-free rate
NASDAQ	National Association of Securities Dealers Automated Quotations
NYSE	New York Stock Exchange
OLS	Ordinary Least Squares
RIC	Refinitiv Instrument Code
S&P	Standard and Poor's
SMB	Small Minus Big
VIF	Variance Inflation Factor
vwret	Value-weighted average return

List of Symbols

$\hat{\beta}_k$	the estimated regression coefficient of the k^{th} independent variable
$\hat{\Omega}$	covariance matrix of factors
$P_{\text{bitcoin},t}$	closing price of Bitcoin at the end of time t
P_t	closing price of stock at the end of time t
R^2	R-squared. Measure of how well the data is fitted by a model
R_f	risk free-rate
R_{it}	return of asset/portfolio i at the time t
R_m	market return
X_i	the dependent variable
d_L	lower critical value
d_U	upper critical value
$\hat{e}$	vector of residuals
m_t	the stock price at the time t
$\hat{\alpha}$	intercepts during the estimation process
ε_{it}	residual component of the regression model for the asset i at the time t
$\bar{\mu}$	the vector representing the means of factor portfolios
μ_{it}	expected return of asset i at the time t
h	factor sensitivity to book-to-market ratio factor
$E(D_{t+\tau})$	expected dividends of the stock during the period $t + \tau$
ESS	explained the sum of squares
F	matrix of factors derived from excess returns in portfolios
K	sample kurtosis
L	quantity of factors incorporated in the model
N	number of portfolios
RSS	residual sum of squares
S	sample skewness
$SE(\hat{\beta}_k)$	estimated standard error (the square root of the estimated variance of the distribution) of $\hat{\beta}_k$
T	count of return observations
TSS	the total sum of squares
b	coefficient of predictor
c	factor sensitivity to investment

n	number of observations
s	factor sensitivity to size factor
v	factor sensitivity to profitability
β	factor sensitivity to market factor
δ	standardized factor loading

1. Introduction

To attract savings towards the equity market, it is imperative to cultivate the confidence of investors. Investors endeavor to designate their savings in a local that presents the utmost return. Nonetheless, they must also consider the risks linked to investment. Considering that most investors are risk-averse, they only invest in securities when they can obtain the best returns possible relative to the risk they take.

On the other hand, investors and market participants now need tools, methods, and models that can assist them in choosing the most optimal portfolio considering the development and growth of marketplaces and instruments for finance, the complexity of financial markets, as well as the specialization of investment. As a result, many theories, models, and techniques for asset pricing and predicting stock returns have grown; these are constantly evolving and continuously shifting. A lot of research that was done until now after the introduction of the Fama-French three-factor model in 1993 shows the popularity and usefulness of this model¹. Fama and French in 1993 found that the three factors of size, book-to-market ratio, and market risk premium have a high power to explain the changes in stock excess returns.

Furthermore, after the emergence of the cryptocurrency market, many investors thought of investing in this market, and the level of progress and investment in this market was such that the market capitalization of this market increased from 11.26 billion dollars in January 2014 to nearly 3000 billion dollars in November 2021 (see [Appendix A, Figure 1](#)). The most popular and traded cryptocurrency is Bitcoin², which has a market capitalization of 813.96 billion dollars as of 5. December 2023, that is 54.39% of the cryptocurrency market (see [Appendix B, Table 1](#)). Bitcoin is the leader of this market, and this has led to plenty of articles that studied the relationship between Bitcoin and other financial markets.

What all past studies have in common is that they calculated the relationship between Bitcoin returns and stock returns by measuring the relationship between Bitcoin returns and stock indices, such as [Nguyen \(2022\)](#), [Matkovskyy and Jalan \(2019\)](#), [Wang et al. \(2022\)](#), [Lopez-Cabarcos et al. \(2021\)](#), and [Mizerka et al. \(2020\)](#). According to my knowledge, only [Cahill et al. \(2020\)](#) and [Cheng et al. \(2019\)](#) have devoted part of their study to measuring this relationship at the Fama and French (F&F) stock portfolio level. In this thesis, I want to measure the correlation between the return of Bitcoin and the excess return of portfolios formed according to Fama-French three-factor model instructions. To be more precise, the study first investigates

¹ [Anuno et al. \(2023\)](#), p. 1.

² [Lopez-Cabarcos \(2021\)](#), p. 1, [Wang et al. \(2022\)](#), p. 4.

whether factors of market risk premium, size, and value have a notable impact on determining the excess return of stock portfolios. Then I compare my results with the results extracted from the literature. So, the study is based on the [Fama and French \(1993\)](#) factor model. After that, I add the return of Bitcoin as an external factor to the model to analyze, whether it has enough power to explain the portfolio excess returns³, how the correlation between these two variables is, and how the adjusted R-squared of the model changes after adding the return of Bitcoin. According to modern portfolio theory, investors should not choose assets solely based on their unique characteristics. Rather, they must assess the relative movements of every asset concerning every other asset. Considering these interactions allows investors to build portfolios with similar expected returns but lower risk compared to portfolios that ignore the relationships between assets⁴. Therefore, the correlation or covariance between the portfolio of stock returns and Bitcoin return should be taken into consideration.

I choose the S&P Composite 1500 as my reference market to implement the Fama and French factors and study the relationship between Bitcoin and stocks. The reason why I chose the S&P Composite 1500 is that this index has never been used for asset pricing models in past studies. Furthermore, this index has enough companies for my research, and it contains a good combination of large-, mid-, and small-cap entities (S&P 500, S&P Mid-cap 400, S&P Small-cap 600), and it covers 90% of the U.S. market capitalization. The launch date of this index is 18th May 1995, which is before the research period in this thesis, so the information on stocks is well available⁵.

In this thesis, stocks are divided into six portfolios according to their size and book-to-market ratio. To analyze the explanatory power of the time series factors in each portfolio, Ordinary Least Squares (OLS) regressions are employed. The reason why I use this method is that for each portfolio I have variables that change over time and the fact is that I analyze each portfolio separately. After running the OLS regressions, I analyze the portfolios jointly as panel data. Panel data is a combination of time series and cross-sectional data⁶. As in my study, there are several portfolios with variables that change over time, so the panel data analysis would be a better model and it is utilized to analyze different factors and their influence on pricing within the cross-section of asset returns over time. The discoveries outlined in this thesis will be valuable for both academics and professionals such as investors, financial institutions, asset management companies, and portfolio and risk managers, aiding them in recognizing crucial

³ [Cahill et al. \(2020\)](#), p. 12.

⁴ [Elton and Gruber \(1997\)](#), p. 1744.

⁵ See [the S&P Composite 1500 factsheet as of 30. November 2023](#).

⁶ [Brooks \(2019\)](#), p. 487.

pricing elements within both the U.S. and cryptocurrency markets. This, in turn, enhances their ability to make more accurate predictions about future equity returns.

The period of the study is July 2003 - December 2022. To make meaningful conclusions, it is necessary to examine asset pricing studies for at least 20 years⁷. But the Bitcoin price data in both Refinitiv (Thomson Reuters) and other databases used in previous studies like [Nguyen \(2022\)](#), [Wang et al. \(2022\)](#), and [Lopez-Cabarcos \(2021\)](#)⁸ is available from the beginning of 2014 onwards. Therefore, I must divide my research into three parts: First, as explained, I will examine the Fama and French three-factor model in the S&P Composite 1500 over twenty years (long term) and check whether the three factors can explain the fluctuations in excess returns of stocks. In the next step, I will perform the same procedure for the period from July 2018 to December 2022 (short term) and check how the results are different in the two periods and finally, I will add the return of Bitcoin. Therefore, one of the limitations of the research is that the review period for the implementation of the asset pricing model and the price of Bitcoin in the second and third phases is shorter than twenty years. To tackle this issue, I use daily intervals instead of monthly. The reason why I started in 2018 is because the number of blockchain-based companies in 2018 is more than in previous years. [Benedetti and Kostovetsky \(2021\)](#) mentioned in their paper that www.coinmarketcap.com is the best database for collecting cryptocurrency data, so, I provide the Bitcoin prices from this website. Refinitiv Eikon is used for collecting the stock market and accounting data, and further calculations regarding factors calculation, portfolio constructions, and statistical tests are done with Stata.

The overall findings of the study show that the market risk premium, size, and value factors can explain the changes in excess return both in daily and monthly intervals. The adjusted R-squared in both models is above 70% and all coefficients are statistically significant. In this thesis, the results of the Fama and French study in 1993 have been proven. After including the return of Bitcoin, I observed that the explanatory power of the model reduced massively to 19.31% and the logarithm of HML became redundant. The coefficients of the other three factors are statistically significant, which means that the logarithm of market risk premium, size, and Bitcoin return can explain the fluctuations in the logarithm of portfolio excess returns, and the logarithm of Bitcoin return has a positive coefficient, which means that Bitcoin diversifies the stock portfolios.

The remainder of this thesis is structured in the following manner. In the second chapter, I discuss the results from past studies. In the next chapter, I provide a theoretical framework and

⁷ [Cochrane \(2005\)](#), p. 287, [Ben Ammar et al. \(2018\)](#), p. 294.

⁸ These databases are: www.coingecko.com, <https://cn.investing.com/>, www.coindesk.com.

introduction to asset pricing models and cryptocurrency, especially Bitcoin. Chapter four provides a brief analysis of the research methodology and introduces the data used in this thesis. In Chapter Five, I construct models for the study, conduct an empirical assessment, and offer quantitative and qualitative analysis. The final chapter, chapter six, concludes the thesis, offering a summary of the entire study. Additionally, I discuss and acknowledge limitations, while suggesting potential areas for future research.

2. Literature Review

2.1. Fama and French Studies in the USA

Previous studies have found that certain variables have a significant impact on stock excess returns. These variables include market risk premium, market capitalization (size), and book-to-market equity ratio (BE/ME). Some of these studies are as follows:

As mentioned earlier, in their research, [Fama and French \(1993\)](#) introduced a three-factor model that incorporated excess market return, company size, and the book-to-market ratio. Their study built upon the findings from [Fama and French \(1992a\)](#), which revealed a significant positive correlation between stock excess returns and both the size and book-to-market ratio, indicating the presence of size and value effects. Fama and French constructed portfolios aimed at capturing risk factors linked to size and BE/ME, and these portfolios effectively capture considerable variability in stock excess returns. One of the noteworthy conclusions from their research is that when stock betas exhibit no correlation with company size, the relationship between beta and stock excess returns remains consistent. This phenomenon diminished the explanatory power of market risk premium, highlighting the significance of the size effect. Moreover, their study underscores that the value effect remains more pronounced than the size effect, as demonstrated by their analysis. In their study, the R^2 of the time-series regressions for each portfolio ranges from 83% to 97%. They assert that a portfolio constructed from market risk premium, size, and book-to-market ratio factors substantially better explains the variations in stock returns than the Capital Asset Pricing Model (CAPM), which has only one factor. According to their findings, the beta of market risk premium alone could account for only 70% of the returns in a diversified portfolio, whereas their three-factor model exhibited a remarkable explanatory power of 95%.

Another study in 2011 by [Kang et al.](#), developed a conditional version of the Conventional Capital Asset Pricing Model (C-CAPM) to investigate the link between macroeconomic factors and equity pricing from 1963 to 2005. They use a unique conditioning variable derived from cointegrated macroeconomic variables, including dividend yield, term spread, default spread,

and short-term interest rate, to predict market excess returns. Their conditional C-CAPM performs impressively, nearly matching Fama and French's three-factor model in explaining the cross-section of stock excess returns, particularly for size and book-to-market sorted portfolios. Their findings support a risk-based interpretation of size and book-to-market effects by showing that value equities are riskier than growth stocks during economic downturns.

[Fama and French in 2015](#) tried to extend their three-factor model and introduced a five-factor model aimed at capturing size, value, profitability, and investment patterns in stock returns. Despite the performance of the factors being relatively insensitive to their definitions, the inclusion of profitability and investment factors renders the value factor redundant. Empirical tests show that the five-factor model can explain a substantial portion of cross-sectional variance in expected returns for various portfolios. However, the study also discusses the difficulty in choosing between different sets of factors and the potential limitations of over-controlling for too many factors. In conclusion, Fama and French's study highlights the importance of the new factors in explaining stock returns and provides insights into the complex interplay between size, value, profitability, and investment patterns in asset pricing models. The high average return of HML stocks could be fully explained by their relationships with the market risk premium ($R_m - R_f$), small minus big (SMB), and, especially, two other factors. This discovery suggests that HML might not be as crucial as previously thought. Another significant result from the study is that the companies that are smaller in size and have a greater book-to-market ratio tend to produce elevated average returns.

In 2020, [Ang and colleagues](#) conducted a similar study on the New York Stock Exchange (NYSE). The results show that the market risk premium and SMB have significant and positive impacts on the changes in stock values. However, the HML factor was found to be unable to significantly explain this variation.

The research paper written by [Bera et al.](#) in 2020 explores the relationship between average stock returns and risk factors using a wavelet multi-scaling approach, allowing for the investigation of this relationship across different time scales. The study uses data from July 1963 to February 2018 collected from the Kenneth R. French website. The data includes various portfolios categorized by size, book-to-market, operating profitability, and investment. Each portfolio is further analyzed at five different time scales. The findings reveal several important insights. First, the relationship between risk factors and average returns varies across different time scales. Second, when assessing the intercepts in asset pricing models, the study indicates that there are abnormal returns in raw-data models, but not in scale-basis models. Third, the study uncovers a period of unexpectedly higher cash flows for big-value portfolios at higher

scales. Fourth, the research challenges the notion that mispricing is a short-term phenomenon, as it appears to have a stronger impact on long-term investment horizons and robust portfolios. Analogous to the F&F results, the study by [Bera et al.](#) demonstrates that groups of small stocks are more volatile than groups of big stocks. Furthermore, the average returns on small stocks are higher than those on large stocks. They demonstrated that while the small-size high-BE/ME portfolio has the highest return, the small-size low-BE/ME portfolio has the highest standard deviation. The market factor is the most effective factor for average returns as measured by its coefficient magnitudes for all time scales and raw data estimates. As the time scale increases, the coefficients of the SMB factor do not increase, except for the small-size high-BE/ME model. Likewise, the coefficients of the HML factor do not increase, except for the large portfolios at high scales. The results show that the explanatory power of the models depends on the scale basis. Finally, this study proved that the HML, SMB, and market risk factors in the five-factor model can significantly explain the variations in the average returns of portfolios.

[Horvath and Wang](#) conducted a study in 2021 for the period January 1990 – March 2020 which assesses the Fama-French model's efficacy on U.S. stock markets during significant events (Dotcom bubble in 1999-2002, Financial crisis in 2007-2010, and Debt crisis in 2009-2013), specifically focusing on the COVID-19 pandemic's impact. In this study, they constructed six value-weighted portfolios based on size and BE/ME, six value-weighted portfolios based on size and investment, and six value-weighted portfolios based on size and operating profitability for each month. Interestingly, they observed a decline in beta values for all five factors before the financial crisis. During this period, there was a notable distinction between the beta values of the market risk premium factor and the other four factors. Rolling beta coefficients display variations, with COVID-19 notably reducing all beta coefficients. According to their findings, R^2 changes a lot during different crises, for example, the R^2 of the regression model increased in financial crisis from 55.35% in July 2007 to 94.83% in June 2008. Notably, the SMB and HML factors were not statistically significant in this study.

Many studies tried to expand the Fama and French models to understand whether more factors can explain the excess return of stocks. In their research in 2019, [Gonzalez and Jareno](#) conducted a comprehensive study comparing twelve different factor models to explain variations in U.S. sector stock returns using the S&P 500 between November 1989 and February 2014. The research built upon the Fama and French three-factor and five-factor models, expanding them by including additional factors such as nominal interest rates, real interest rates, and expected inflation rates. The results of the three-factor model show that the coefficients of market risk and value are significantly positive, however, it is significantly negative for the size

factor. The utility sector consistently displayed the lowest explanatory power across all models, quantiles, and periods. Moreover, the study reveals that by adding more factors the explanatory power of the models increases. Another important result obtained in this study, and which can affect the results of my research is that the explanatory power of variables changes during recession and expansion.

[Ugurlu-Yildirim and Sendeniz-Yuncu's](#) paper (2021) explores the impact of institutional investors on asset pricing models in the NYSE, American Stock Exchange (AMEX), and NASDAQ. Their study used time-series regression analysis, utilizing information from non-financial companies that were listed between January 1980 and December 2016. The results show that the Institutional minus Individual (IMI) component, added to the Fama-French five-factor model, provides a more comprehensive explanation for typical return differences, particularly in portfolios with different amounts of institutional ownership. Additional research reveals that substantial common variances in returns are captured by the market factor and portfolios that are built to replicate size, book-to-market equity, and institutional ownership factors. By showing that size, book-to-market, and momentum are good proxies for common risk elements in stock returns and by adding institutional ownership as an additional representation of a shared risk element, the research validates the conclusions of [Fama and French \(1993\)](#). In alignment with what [Fama and French \(1993\)](#) discovered, the adjusted R^2 values, which indicate how well the model explains the data, are only approximately 70 percent for portfolios consisting of larger stocks with low book-to-market equity ratios.

2.2. Fama and French Studies out of the USA

In addition to U.S. financial markets, many researchers examined the effectiveness of the Fama and French models in markets outside the United States. Most of these experiments showed different results from what was done in the U.S. markets. In 2019, [Feyyaz et al.](#) examined the applicability of Fama and French's Five Factor Asset Pricing Model (F&F5) on firms listed in the Istanbul Stock Market Sustainability (ISMS) Index. The research found that while higher market value correlated with greater returns, the influence of the book-to-market ratio is not statistically significant. Ultimately, the study concluded that there's limited evidence supporting F&F5's validity for the listed companies, aligning with prior research.

[Gonzalez-Sanchez \(2022\)](#) shows that factorial asset pricing models generally don't work very well across a set of twenty-six emerging equity markets.

The study of [Mosoeu and Kodongo](#) in 2022 in emerging and developed equity markets reveals a notable weakness in the applicability of the Fama-French five-factor model based on the

Gibbons-Ross-Shanken (GRS) tests, particularly when applied to country-specific and geographically diversified portfolios.

2.3. Cryptocurrency and Equity Market Studies

The idea of adding the Bitcoin return as a fourth factor to the F&F3 model in this thesis comes from the following two studies by [Cahill et al. \(2020\)](#) and [Cheng et al. \(2019\)](#):

[Cahill et al. \(2020\)](#) explored how publicly listed blockchain-related stocks responded to announcements about blockchain in the USA using 8-K reports. In a global study encompassing 713 firm announcements, they found that, on average, there was an abnormal return of about 5% on the day of the announcement. The research questions in this study are whether Bitcoin can explain abnormal returns and whether some factors cause these abnormal returns. Upon introducing Bitcoin returns as an additional factor to Fama and French five-factor model to elucidate portfolio returns, the results revealed positive, albeit insignificant, Bitcoin returns across the entire sample. The low R^2 of the regression indicates that the factors do not capture completely the variations in portfolio excess return. The main goal of another study by [Cheng et al. \(2019\)](#), which implements the Fama-French three-factor model, was to analyze how the stock prices react to initial 8-K disclosures categorizing the companies in Speculative and Existing. The authors expect a company to be exposed to blockchain technology if its stock returns move with Bitcoin returns. The results show that Bitcoin has a significantly positive coefficient, both in the case of Speculative and Existing stocks. The coefficients are respectively 0.04 and 0.09. In addition, the coefficient of SMB, HML, and market risk premium in this study are statistically significant.

As stated in the introduction section, many other studies have investigated the impact of digital currencies on stock returns or vice versa using stock market indices. For instance, [Gil-Alana et al. \(2020\)](#) determined that the world of cryptocurrencies is turning out to be a more significant factor in financial markets than initially thought. This is mainly because it provides investors with a valuable diversification option, given its limited correlation with traditional assets. This lack of correlations between Bitcoin returns and the stock market indexes means that Bitcoin plays either a hedging or diversifying role for investors in their portfolios depending on the sign of the coefficient⁹. Furthermore, [Baur et al. \(2018\)](#) discovered that the return of Bitcoin does not have a significant correlation with conventional assets like stocks, bonds, and commodities. This holds true in both stable market conditions and during financial crises in the USA. The investigation into Bitcoin account transaction data reveals that Bitcoin is predominantly utilized

⁹ [Chan et al. \(2019\)](#), p. 113, [Caferra and Vidal-Tomas \(2021\)](#), p. 8, [Gil-Alana \(2020\)](#), p. 9.

as a speculative investment rather than being embraced as an alternative currency or medium of exchange.

In her paper, [Dyhrberg \(2016\)](#) examined how Bitcoin returns relate to the Financial Times Stock Exchange (FTSE) index using an asymmetric GARCH model. She aimed to analyze whether Bitcoin can be a hedge against the companies. The research investigates the potential of Bitcoin to act as a hedge alongside gold, effectively mitigating specific market risks, particularly against the FTSE. In this study, Bitcoin was uncorrelated with the stocks in the FTSE Index. In conclusion, the findings indicate that Bitcoin does possess important hedging capabilities against the FTSE Index.

Another study by [Chan et al.](#), which was conducted in 2019, showed that there is a negative correlation between monthly Bitcoin returns and five indices, namely the Euro STOXX, NIKKEI, Shanghai A-Share, S&P 500, and the TSX Index. Therefore, Bitcoin has the role of hedger under monthly data frequency and weak hedger under weekly and daily data frequency. For the monthly frequency, the correlations between Bitcoin return and the S&P 500, NIKKEI, Shanghai A-Share, TSX, and STOXX are respectively -0.0202, -.0089, -0.0054, -.0228, and -0.0143.

In a comprehensive study, [Huang et al. \(2021\)](#) investigated whether Bitcoin could serve as a safe haven during the COVID-19 bear market in five major economies. The findings indicate that, in Australia, Bitcoin is primarily an effective portfolio diversifier for local stocks, while in Canada, it acts as a weak safe haven. In Europe, the UK, and the US, Bitcoin is identified as a weak safe haven.

In a study by [Mariana et al. \(2021\)](#), the suitability of Bitcoin and Ethereum as short-term safe havens for stocks was examined. The findings reveal that the two major cryptocurrencies exhibit characteristics of short-term safe-havens, particularly evident during the pandemic where their daily returns tend to inversely correlate with those of the S&P500. The results of the study consistently support these safe-haven attributes and suggest that Ethereum might even be a more effective safe haven than Bitcoin. Throughout the pandemic, the correlation between S&P500 and Bitcoin was -0.38, while for Ethereum, it was -0.37. These findings suggest that both cryptocurrencies show early indications of being potential safe havens during periods of stock market uncertainty. Additionally, the study notes that the dynamic correlations between S&P500 and Bitcoin before the pandemic were not consistently negative, varying between -0.07 and 0.10, with a median of -0.0047.

After reviewing past studies, some researchers believe that the return of Bitcoin does not have a significant correlation with the fluctuations of stock returns. Another group believes that there

is a significant relationship between these two variables. In this thesis, I will discuss whether the return of Bitcoin is related to the excess return of the portfolios formed from the companies in the S&P Composite 1500 based on the Fama and French three-factor model. More precisely I examine whether the coefficient of Bitcoin is significant and how the four factors can jointly explain the variations in stock excess returns.

3. Theoretical Framework

3.1. Key Asset Pricing Theories

Many studies were carried out on the behavior of stock returns during the 1900s, which resulted in the creation of important ideas. The most crucial among them is the Capital Asset Pricing Model (CAPM). This model faced criticisms, some of which were articulated in studies by [Banz \(1981\)](#), [Basu \(1983\)](#), [De Bondt and Thaler \(1985\)](#), and [Bhandari \(1988\)](#). However, it wasn't until [Fama and French \(1993\)](#) that confirmation emerged.

The following reference is made to some of the asset pricing models.

3.1.1. Capital Asset Pricing Model (CAPM)

The most used model in the field of capital markets for measuring risk and return is the Capital Asset Pricing Model (CAPM). This model comprises a set of predictions about the expected returns of risky assets in equilibrium, which was simultaneously and independently developed twelve years after [Markowitz's \(1952\)](#) pioneering work, by [Sharpe \(1964\)](#), and [Lintner \(1965\)](#). By combining borrowing or lending against a portfolio with a certain amount of risk, the Sharpe model adjusts for risk. This portfolio, which comprises all stocks proportionately weighted by their coefficients, is the market portfolio, according to Sharpe. There are two presumptions in this model: 1) In a symmetric environment, beta reflects risk; that is, an investor will earn returns commensurate with the risk they assume. 2) In other words, beta is the only explanatory factor for returns.

3.1.2. Fama-French Three-Factor Model

The model proposed by [Fama and French \(1993\)](#) which is appropriate for the panel data, is as follows:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + s_iSMB_t + h_iHML_t + \varepsilon_{it}, \quad (3.1)$$

where:

$R_{it} - R_{ft}$: Excess stock/portfolio return. R_{it} is the stock/portfolio return at the time t and R_{ft} is the risk-free rate at the time t .

$R_{mt} - R_{ft}$: Excess market return, also known as market risk premium. R_{mt} is the market return of the S&P Composite 1500 at the time t .

SMB_t : The size factor is the difference between the average returns of portfolios of large companies and the average returns of portfolios of small companies.

HML_t : The book-to-market ratio factor is equal to the difference between the average returns of stocks with high book-to-market ratios (value) and the average returns of stocks with low book-to-market ratios (growth).

α_i is the intercept of the model which shows how much of the portfolio's excess return variations are not explained by the independent variables.

β_i, s_i, h_i are, in order, factor sensitivities to market, size, and book-to-market factors.

ε_{it} : The residual of the regression.

For the time-series regressions for each portfolio, I use the following model:

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + s_iSMB + h_iHML + \varepsilon_i. \quad (3.2)$$

As it is known, the difference between equations 3.1 and 3.2 is in index t . When we have panel data, regression will be done every month or every day, but in time series data, we only have a general regression for the entire period.

The Dividend Discount Model explains the rationale behind linking the book-to-market ratio to average stock returns. According to this approach, a stock's market value is equal to the present value of its anticipated dividends over a given period, as expressed below:

$$M_t = \frac{\sum_{\tau=1}^{\infty} E(D_{t+\tau})}{(1+r)^\tau}. \quad (3.3)$$

In this equation, M_t represents the stock price at time t , $E(D_{t+\tau})$ is the expected dividends of the stock during the period $t + \tau$, and r is the long-term average return or, more precisely, the expected internal rate of return on dividends. After dividing the equation 3.3 by book value at time t (B_t) and inverting the fraction, the following formula is obtained:

$$\frac{B_t}{M_t} = \frac{B_t(1+r)^\tau}{\sum_{\tau=1}^{\infty} E(D_{t+\tau})}. \quad (3.4)$$

According to equation 3.4, there is a relationship between the book-to-market ratio of stocks and their expected returns¹⁰. This is the main reason for including the HML factor as a variable that might explain the variations in stock excess return. According to equation 3.4, stocks with higher BE/ME have higher returns.

Fama and French also mentioned in their papers ([1992a](#) and [1993](#)) that size has a relation with profitability. They checked all the companies in their sample and concluded that the small firms have lower earnings on assets than big firms, therefore they came up with the hypothesis that the size should have a relation with the stock returns¹¹.

3.2. Factor-Portfolio Construction

In the literature (e.g., [Fama and French \(1993\)](#), [Fama and French \(2015\)](#), [Mosoeu and Kodongo \(2022\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), [Horvath and Wang \(2021\)](#), [Bera et al. \(2020\)](#)), there are two widely used methods for constructing factor portfolios, as outlined in the following:

1) The Method of Forming Factors as a 2×2 Matrix:

For this purpose, companies are first ranked based on various degrees of size and the ratio of book value to the market value of equity. Then, using the median of all stocks in the S&P Composite 1500, they are divided into two groups. Subsequently, a 2×2 matrix is formed, where one component represents the size of the company, namely small and big companies, and the second component represents the ratio of book value to market value of equity, namely low book-to-market, and high book-to-market. As a result, from the intersection of these groups, four portfolios are created. Finally, the formation of factors is addressed.

2) The Method of Forming Factors as a 2×3 Matrix:

For this purpose, companies are categorized based on various degrees of size and the ratio of book value to market value. They are divided into two sizes using the median of stock market capitalizations and further divided into three sections using the 30th percentile and 70th percentile of the ratio of book value to market value. Here are also the portfolios sorted based on the median and percentile of all stocks in the S&P Composite 1500. Subsequently, a 2×3 matrix is formed where one component represents company size, and the second component represents the ratio of book equity to market equity. The intersection of all groups yields six portfolios, and finally, the factors are calculated.

¹⁰ [Fama and French \(2006\), p. 492.](#)

¹¹ [Fama and French \(1993\), p. 8.](#)

Regarding diversification, the portfolios, which use only median, are more diversified in comparison to the 2×3 version because the 2×2 version considers all the stocks of HML, but the 2×3 version excludes 40% of stocks. Based on the F&F article¹², there are very few differences between the results of these two methods, and since in many past studies and on the Kenneth R. French website¹³, the 2×3 method is used, I constructed 2×3 sorts on size using the median and BE/ME using the 30th and 70th percentiles as breakpoints. The other method (2×2) is not available on this website.

There are also more portfolio construction methods, such as 5×5 and 10×10. However, the two methods explained earlier were mostly used in past studies.

3.3. Bitcoin and Cryptocurrency

It was around 2008 that many individual and institutional investors started to get acquainted with cryptocurrency¹⁴. Bitcoin is a digital currency and, therefore, has no physical existence. The Bitcoin blockchain is a database of all Bitcoin transactions. This database also contains information about the production of new Bitcoins. From the beginning, there has been much debate about the nature of Bitcoin, whether it can truly be considered a type of currency, a form of capital, or a commodity, or if Bitcoin has no intrinsic value.

After several years, the acceptability of Bitcoin and digital currencies in general has reached such a level that the most well-known stock exchanges of the world created various indices for them, and even their derivative instruments are being traded. Some of these indices are the S&P Bitcoin Index, the FTSE Bitcoin Index, the S&P CME Bitcoin Futures Index, the National Association of Securities Dealers Automated Quotations (NASDAQ) Crypto Index, etc. Cryptocurrencies represent a novel form of assets¹⁵ that are actively traded across various platforms. Also, on 10.01.2024, the U.S. Securities and Exchange Commission (SEC) approved Bitcoin tradable funds¹⁶, which is formally a confirmation of the acceptability of Bitcoin and digital currencies in the financial world. However, they differ from stocks in terms of their unique characteristics. For example, the price volatility of cryptocurrencies distinguishes them from traditional financial assets¹⁷. Cryptocurrencies are categorized under the heading of digital assets, which, unlike stocks and bonds, have no intrinsic value¹⁸.

¹² Fama and French (2015), p. 19.

¹³ See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html [last accessed on 23.02.2024].

¹⁴ See Nakamoto, 2008.

¹⁵ Gil-Alana et al. (2020), p. 1.

¹⁶ See <https://www.sec.gov/news/statement/gensler-statement-spot-Bitcoin-011023> [last accessed on 23.02.2024].

¹⁷ Wang (2022), p. 1

¹⁸ Glas (2022), p. 47, Goutte et al. (2019), p. 70.

In March 2014, the [Internal Revenue Service \(IRS\)](#) in the United States declared that all virtual currencies, including Bitcoin, are subject to asset taxation and are not considered actual currency. Additionally, any gains or losses resulting from holding Bitcoin as an investment are treated as capital gains and losses, whereas if held as inventory, they will be considered ordinary gains and losses¹⁹, and these gains and losses are reflected in the financial statements of the firms. When adding the returns of Bitcoin to my model, I select companies from the S&P Composite 1500 that are somehow related to blockchain technology or cryptocurrencies, especially Bitcoin. This relation could be in product or service offerings, company strategies, way of transactions and payments, or an item in the firm's balance sheet or investment portfolio²⁰. Because investors may use pricing data from the cryptocurrency market to predict the stock prices of firms that have implemented blockchain technology or hold cryptocurrencies on their balance sheets, these companies are now more susceptible to swings in the market. However, as seen by the jumpy nature of their returns, cryptocurrencies have experienced regular and significant price discontinuities²¹. Therefore, it is expected that there is a correlation between the price of Bitcoin and the stock price of companies related to blockchain and cryptocurrencies. Furthermore, the gains and losses from holding cryptocurrencies or activity based on blockchain technology affect the profitability of the firms which can be subject to changes in stock prices.

When investors decide to invest in stocks, they simultaneously consider a set of variables and financial and non-financial factors. Decision-makers, with an awareness of the factors influencing stock performance, can more accurately assess the behavior of stock prices and consequently make more informed decisions.

Adding the return of Bitcoin to the model has a particular role in financial analysis and managing risk. Here are two possible reasons for including the returns of Bitcoin in the model:

- 1- Diversification Analysis: Spreading their investments helps investors reduce risk. Analysts can assess whether incorporating Bitcoin into a portfolio of stocks from the S&P Composite 1500 can provide benefits in terms of diversification by including the return of Bitcoin into the model. Compared to traditional assets like stocks and bonds, Bitcoin acts as a portfolio diversifier when the coefficient is noticeably positive but

¹⁹ The information can be found on the official website of the IRS: <https://www.irs.gov/individuals/international-taxpayers/frequently-asked-questions-on-virtual-currency-transactions> [last accessed on 23.02.2024].

²⁰ Xu et al. (2022), p. 1.

²¹ Xu et al. (2022), p. 1.

weak²². If the coefficient is not substantially different from zero, Bitcoin may be considered a limited safe haven against stocks and bonds during financial crises²³.

- 2- Hedging Strategy: By examining the model's sensitivity to Bitcoin returns, we can determine whether the cryptocurrency can serve as a hedge against specific risks in the S&P Composite 1500. Should Bitcoin exhibit a robust negative coefficient, it may serve as a hedge, meaning it can rise in value when more traditional assets fall in value.²⁴ Additionally, an insignificant coefficient indicates a weak hedge²⁵.

4. Data and Methodology

4.1. Data

I acquired all the market and accounting data from Refinitiv Eikon (formerly Thomson Reuters) from January 2002 to December 2022 (I need yearly accounting data from 2002, but the monthly stock price from July 2003). I use Stata for data processing, portfolio construction, and statistics tests. The reference index is S&P Composite 1500. This index contains Large-cap, Mid-cap, and small-cap companies. The number of constituents is 1505, and the launch date of the index was on 18th May 1995²⁶.

Both the Bitcoin price and the stock closing price for the period from July 2018 to December 2022 are daily when I integrate the Bitcoin and Fama-French three-factor model, as other researchers such as [Matkovskyy and Jalan \(2019\)](#), [Wang et al. \(2022\)](#), [Lopez-Cabarcos et al. \(2021\)](#), and [Mizerka et al. \(2020\)](#) did in their studies. I use daily data because the period is less than twenty years, and this offers increased flexibility in evaluating Fama and French factors within shorter periods that would be challenging with monthly data²⁷.

According to [Fama and French \(1992a\)](#), [Hou et al. \(2015a\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), [Walkshäusl \(2021\)](#), and [Huang et al. \(2021\)](#), I include all nonfinancial companies in my analysis while deliberately leaving out financial firms (with 4-digit SIC codes 6000 - 6999)²⁸. This choice was made because the substantial debt levels typically found in financial firms might not carry the same implications as they do for nonfinancial firms. In the case of nonfinancial companies, high leverage often signifies financial troubles and distress.

²² [Huang et al. \(2021\), p.5](#), [Chan et al. \(2019\), p. 111](#).

²³ [Huang et al. \(2021\), p.5](#).

²⁴ [Huang et al. \(2021\), p.5](#), [Chan et al. \(2019\), p. 111](#).

²⁵ [Chan et al. \(2019\), p. 111](#).

²⁶ [S&P Composite 1500 factsheet as of 30. November 2023](#).

²⁷ [Faff \(2003\), p. 312](#).

²⁸ [Yu \(2022\), p. 2](#), [Nichols \(2017\), p. 1388](#).

Furthermore, I exclude the utility firms (with 4-digit SIC codes 4900 - 4999)²⁹, because these firms are highly regulated³⁰. To minimize the impact of unwanted data fluctuations and potential skewing or outliers³¹, I adjust all the variables by capping extreme values at their 5th and 95th percentiles (winsorization)³².

To find the blockchain and cryptocurrency-related companies, I used the 10-K filings (disclosures) from the SEC's Electronic Data Gathering, Analysis and Retrieval (EDGAR)³³. The keywords in my research were "Bitcoin", "Cryptocurrency", and "Blockchain"³⁴. I use 10-K filings instead of 8-K filings, while 10-K filings give complete information regarding the financial statements of firms and the past activities and investments as well as the projects and decisions in the future.

The reason why I started the study period at this stage in 2018 is that many companies have shown their interest in blockchain technology since 2018³⁵. After searching through the 10-K reports, it was found that the number of companies that have incorporated blockchain technology or cryptocurrencies before 2018 is too small.

The total number of companies in each year that can be investigated in the period from 2018 and are related to blockchain and cryptocurrency is 29. These companies can be seen in [Appendix C, Table 1](#).

4.2. Variables and their Operational and Conceptual Definitions in the Research

In this research, the dependent variable is the portfolio excess return, and the independent variables are market excess return (market risk premium), company size, book-to-market ratio, and Bitcoin return.

R_{it} : The return of stock or portfolio i in period t

To get the continuously compounded returns for the stocks, I calculate the natural logarithmic changes in value³⁶ according to equation 4.1. This calculation is based on the closing stock price at the end of the current month/day compared to those at the end of the previous

²⁹ The utility industry consistently demonstrates the least ability to explain variations (as measured by Adj. R^2) in the Fama-French models compared to other industries ([Gonzalez and Jareno \(2019\), p. 188](#)).

³⁰ [Yu \(2022\), p. 2](#).

³¹ [Yu \(2022\), p. 2](#), [Huang et al. \(2021\), p. 2](#).

³² [Bank and Insam \(2021\), p. 3](#).

³³ See <https://www.sec.gov/edgar/searchedgar/companysearch> [last accessed on 23.02.2024].

³⁴ [Cheng et al. \(2019\), p. 5902](#), [Cahill et al. \(2020\), p. 8](#).

³⁵ [Cahill et al. \(2020\), p. 3-4](#), [Xu et al. \(2022\), p. 3](#).

³⁶ [Gonzalez and Jareno, \(2019\), p. 189](#), [Curto and Serrasqueiro \(2022\), p. 3](#), [Chatzitzisi et al. \(2021\), p. 829](#), [Pati et al. \(2019\), p. 339](#), [Bertone et al. \(2015\), p. 74](#).

month/day. Because the closing price has already been modified to account for dividends and stock splits, there is no need to factor in dividends and splits when calculating the returns.

$$R_{it} = 100 \times [\ln(P_{it}) - \ln(P_{it-1})], \quad (4.1)$$

where P_{it} is the closing price for each stock at the end of the month t .

R_{ft} : Risk-free rate of return

The one-month U.S. Treasury bill rate with the Refinitiv Instrument Code (RIC = “US1MFRB=RR”) is considered the risk-free rate of return³⁷.

$R_{it} - R_{ft}$

It represents the company's excess return (the difference between the company's or portfolio's return rate and the risk-free rate of return). This excess return is related to the following three factors in this study. Essentially, these three factors are the main independent variables in the model.

First factor: $R_{mt} - R_{ft}$

The market risk premium is the same as the factor presented in the CAPM model and is referred to as the market factor. It represents the difference between the market portfolio's return rate and the risk-free rate in each period. In this context, R_m represents the market return rate, and R_f represents the risk-free rate. The market return is the monthly/daily return of the S&P Composite 1500 index. The RIC of the S&P Composite 1500 is “SPSUP”.

β : Beta indicates the sensitivity of security returns to market portfolio return fluctuations and is derived from the division of the covariance of security returns with market portfolio returns by the variance of market portfolio returns.

$$\beta_{it} = \frac{\text{Cov}(R_{it}, R_{mt})}{\text{Var}(R_{mt})}. \quad (4.2)$$

Second factor: Size (SMB: Small Minus Big)

This indicates the difference between the average monthly/daily return of portfolios consisting of small-sized stocks and the average monthly/daily return of portfolios consisting of big-sized

³⁷ Fama and French (1993, 2015, 2017), Bera et al. (2020), p. 416, Ugurlu-Yildirim and Sendeniz-Yuncu (2021), p. 5.

stocks, which is referred to as the size factor. Based on theoretical foundations and previous studies like [Dang et al. \(2018\)](#), various criteria are considered for determining the size of a company. These criteria include asset value, sales volume, and market capitalization. In this study, the company's size will be determined by the product of the stock closing price and the number of shares held by shareholders³⁸. Since the companies need to be categorized into big and small firms according to the model's requirements, Fama and French's research approach is followed, using the median criterion of the size of sample companies in the S&P Composite 1500 for this categorization³⁹. In other words, if a company's size is smaller than this criterion, it falls into the category of small companies, and if it is larger, it falls into the category of big companies.

In various studies⁴⁰, the researchers rank at the end of each June year t all the stocks on size using the market capitalization for June of the year t . I follow the same process.

The size factor calculation formula is as follows:

$$SMB_{B/M} = \frac{(\frac{S}{L} + \frac{S}{M} + \frac{S}{H})}{3} - \frac{(\frac{B}{L} + \frac{B}{M} + \frac{B}{H})}{3}. \quad (4.3)$$

S/L : Represents the value-weighted average return of companies with small sizes and low book-to-market ratios.

S/M : Represents the value-weighted average return of companies with small sizes and medium book-to-market ratios.

S/H : Represents the value-weighted average return of companies with small sizes and high book-to-market ratios.

B/L : Represents the value-weighted average return of companies with big sizes and low book-to-market ratios.

B/M : Represents the value-weighted average return of companies with big sizes and medium book-to-market ratios.

B/H : Represents the value-weighted average return of companies with big sizes and high book-to-market ratios.

³⁸ [F&F \(1993\), p. 8](#), [F&F \(1995\), p.131](#), [F&F \(2015\), p. 5](#), [Mosoeu and Kodongo \(2022\), p. 58](#), [Gonzalez-Sanchez \(2022\), p. 1](#).

³⁹ [F&F \(1993\), p. 8](#), [F&F \(1995\), p. 133](#), [F&F \(2015\), p. 9](#), [Bera et al. \(2020\), p. 417](#), [Mosoeu and Kodongo \(2022\), p. 58](#).

⁴⁰ [F&F \(1995\)](#), [F&F \(1993\), p. 8](#), [Lam \(2002\), p. 166](#), [F&F \(2015\), p. 34](#), [Rath and Durand \(2015\), p. 140](#), [Kenneth R. French database](#).

Third factor: Value (HML: High BE/ME Minus Low BE/ME)

Fama and French introduce book common equity (BE) as the book value of stockholders' equity, plus balance-sheet deferred taxes and investment tax credit, minus the book value of the preferred stocks⁴¹. However, according to the update on the Kenneth R. French website, I exclude the deferred taxes and investment tax credits⁴². If redemption value accordingly is available, I use it to estimate the value of preferred stock⁴³. The book-to-market equity ratio is book common equity for the fiscal year ending in calendar year $t - 1$, divided by market equity at the end of December of $t - 1$.⁴⁴ I do not use negative-BE firms⁴⁵. As I mentioned in the SMB section, the previous studies lined up the financial data from the end of each fiscal year in the previous calendar year with the returns from July of the current year to June of the next year. This approach allows for at least a six-month gap⁴⁶ between the fiscal year's end and when testing the returns, which ensures that the stock market has accurately mirrored the accounting information in annual reports like 10-K. This practice helps prevent a forward-looking approach⁴⁷.

The HML (High Minus Low) portfolio is designed to reflect the risk associated with the book-to-market equity factor in investment returns. It is calculated by taking the monthly/daily average returns of two groups of companies, those with high book-to-market ratios (S/H and B/H) and those with low book-to-market ratios (S/L and B/L). The difference between the average returns of these two groups is what makes up the HML portfolio. Based on the research background and following the approach of Fama and French, companies are sorted into three categories: low, medium, and high, based on this ratio. Accordingly, 30% of companies are classified as low, 40% as medium, and 30% as high based on their ranking from low to high. The important point is that the high and low book-to-market portfolios have companies with similar average sizes. This means that the size factor's influence on returns is minimized, allowing me to focus on how firms with high and low book-to-market perform differently in terms of returns. Therefore, the purpose of grouping and creating portfolios based on two

⁴¹ Fama and French (1993), p. 8, Aretz et al. (2010), p. 1386, Hou et al. (2019), p. 3.

⁴² The explanation on the Kenneth R. French database: "Because of changes in the treatment of deferred taxes described in FASB 109, files produced after August 2016 no longer add Deferred Taxes and Investment Tax Credit to BE for fiscal years ending in 1993 or later."
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html [last accessed on 23.02.2024].

⁴³ See Kenneth R. French database.

⁴⁴ F&F (1995), p. 133, Lam (2002), p. 166, F&F (2015), p. 34, Boons (2016), p. 501, Hou et al. (2019), p. 4.

⁴⁵ F&F (1995), p. 133, Davis et al. (2000), p. 393, Mosoeu and Kodongo (2020), p. 59, Pati et al. (2019), p. 339, Hou et al. (2015a), p. 659, Boons (2016), p. 494.

⁴⁶ Noh and Zhou (2022), p. 17, F&F (1992a), p. 429, Lam (2002), p. 166, Cheng and Wu (2010), p. 530, Hou et al. (2015a), p. 660.

⁴⁷ Noh and Zhou (2022), p. 17

factors, size, and value, is to minimize the impact of one factor on returns while analyzing the other factor. Equation 4.4 is the formula of HML calculation.

$$HML = \frac{(\frac{S}{H} + \frac{B}{H})}{2} - \frac{(\frac{S}{L} + \frac{B}{L})}{2}. \quad (4.4)$$

Several studies⁴⁸ suggested that all companies with fiscal years other than December should be excluded from the sample to facilitate the calculation of risk factors and the formation of portfolios. After extracting the required data, I realized that many of the companies in the S&P 1500 have fiscal years other than the end of December. So, I could not remove these companies from my sample because my final sample will include a small number of companies that will not be suitable for conducting statistical tests, and wrong results may be obtained. Therefore, I implemented the same method as Fama and French, and other researchers, who did not exclude the companies with fiscal years other than the end of December from the sample.

There are two methods for calculating the return of each size-value portfolio, equally weighted and value-weighted. In this thesis, the second method will be used. Fama and French mentioned in their paper in 2015 that value-weighted portfolios decrease the volatility of the factors in asset pricing. Additionally, this process obtains the size and value effects, which aligns with the investment prospects⁴⁹.

After providing the daily closing price of Bitcoin, I calculate the return as follows:

$$R_{bitcoin,t} = 100 \times [\ln (P_{bitcoin,t}) - \ln (P_{bitcoin,t-1})], \quad (4.5)$$

where $P_{bitcoin,t}$ is the closing value (price) for Bitcoin at the end of day t .

The potential variation in the correlation between Bitcoin and stock markets implies the unsuitability of employing conventional linear models⁵⁰. As OLS is appropriate for linear relations, one option is to make some transformations in the variables to make the nonlinear relation linear. One of the methods is using logarithms of variables to transform them into linear relations so we can use OLS regression models⁵¹. For example, [Gil-Alana et al. \(2020\)](#) used the

⁴⁸ [Walkshäusl \(2021\)](#), p. 254, [Nichols et al. \(2017\)](#), p. 1388.

⁴⁹ [Fama and French \(1993\)](#), p. 10.

⁵⁰ [Bouri et al. \(2020\)](#), p. 157, [Tong et al \(2022\)](#), p. 1.

⁵¹ [Clements and Preve \(2021\)](#), p. 4, [Kamberi et al. \(2022\)](#), p. 3.

logarithm of Bitcoin and stock market returns in the OLS model⁵². Furthermore, they mention, that Bitcoin and S&P both have linear time trends⁵³. Equation 4.6 is what I use for testing the joint explanatory power of the four variables in a variation of portfolios' excess returns. Additionally, as Bitcoin and stock have different levels of price volatility it is necessary to use the logarithm of the variables as Cahill et al. did for the size and Bitcoin return⁵⁴. The extended Fama-French three-factor model for the panel data is as follows:

$$\log(R_{it} - R_{ft}) = \alpha_i + \beta_i(\log(R_{mt} - R_{ft})) + s_i \log(SMB_t) + h_i \log(HML_t) + b_i \log(R_{bitcoin,t}) + \varepsilon_{it}. \quad (4.6)$$

4.3. Research Questions and Hypotheses

After reviewing the research literature and the results extracted from it, the questions and hypotheses of this research can be stated as follows:

Research Questions

- 1- Can the risk factors (market, size, and book-to-market ratio) collectively explain and significantly capture the monthly stock/portfolio excess returns within the S&P Composite 1500 from July 2003 to December 2022?
- 2- Can the risk factors (market, size, and book-to-market ratio) collectively explain and significantly capture the daily stock/portfolio excess returns within the S&P Composite 1500 from July 2018 to December 2022?
- 3- Which portfolio can achieve a higher average return in both periods?
- 4- Can the return of Bitcoin and the other three factors significantly explain the variations in stock/portfolio excess returns within the Fama and French three-factor model from July 2018 to December 2022?

Research Hypotheses

- 1- The risk factors of market, size, and book-to-market ratio can collectively explain and significantly capture the monthly stock/portfolio excess returns within the S&P Composite 1500 exchange from July 2003 to December 2022.

⁵² [Gil-Alana et al. \(2020\), p. 7-8.](#)

⁵³ [Gil-Alana et al. \(2020\), p. 5.](#)

⁵⁴ [Cahill et al.. \(2020\), p. 6 & 8.](#)

- 2- The risk factors of market, size, and book-to-market ratio can collectively explain and significantly capture the daily stock/portfolio excess returns within the S&P Composite 1500 exchange from July 2018 to December 2022.
- 3- According to studies by Fama and French (1992a, 1995, 2006, 2015), firms with smaller sizes and higher book-to-market ratios can generate higher average returns among other portfolios.
- 4- There is a significantly weak relationship between Bitcoin and excess stock returns. The other three factors have still significant coefficients and the adjusted R^2 decreases within the Fama and French three-factor model.⁵⁵

4.4. Methodology

4.4.1. Descriptive Statistics

In this section, to gain a comprehensive overview of the research variables, some characteristics related to them, including their values such as minimum, maximum, mean, standard deviation, and t-statistics, are summarized.

4.4.2. Data Analysis

In 2015, Fama and French proposed the idea that a reliable model for forecasting expected returns should ideally yield an intercept, often referred to as alpha, that's close to zero. So, when I introduce new factors into the model and notice a significant drop in this alpha value (bringing it closer to zero), it's a sign that these extra factors are improving the model's capacity to make sense of returns of the U.S. economy. Simply put, a smaller alpha suggests that the model performs better when these new factors are included in the analysis. This means that the model effectively explains the returns, leaving no unexplained portion⁵⁶. In my thesis, I examine the performance of the asset pricing models using a statistical test called the Gibbon, Ross and Shanken (GRS) test, similar to the approach used by [Fama and French \(2015\)](#), [Aretz et al. \(2010\)](#), [Kang et al. \(2011\)](#), [Cakici \(2014\)](#), [Bank and Insam \(2021\)](#), [Skocir and Loncarski \(2018\)](#), [Gonzalez and Jareno \(2019\)](#), [Pati et al. \(2019\)](#), [Mosoeu and Kodongo \(2022\)](#), and [Bera et al. \(2020\)](#).

The GRS test, in the realm of finance, is essentially a statistical test known as an F-test. It helps me determine whether all the alpha values (which come from a series of regression analyses

⁵⁵ This is based on the results of [Cheng et al. \(2019\)](#), p. 5910.

⁵⁶ [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), p. 12.

conducted over time) are significantly equal to zero or not. The null hypothesis is stated as $H_0: \alpha = 0$ ⁵⁷, this implies that the starting points (intercepts) for the investment portfolios collectively amount to zero⁵⁸. If the null hypothesis is not accepted, the intercept is statistically significantly different from zero, and the model has cross-sectional pricing errors. To run this test, we do not need to check whether the error terms are normally distributed⁵⁹.

According to [Mosoeu and Kodongo \(2022\)](#), the GRS test statistics can be expressed using the following equation:

$$GRS = \left(\frac{T}{N}\right) \left(\frac{T-N-L}{T-L-1}\right) \left[\frac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{\mu}' \hat{\Omega}^{-1} \bar{\mu}}\right] \sim F(N, T - N - L), \quad (4.7)$$

where:

T : Represents the count of return observations.

N : Denotes the number of portfolios.

L : Signifies the number of factors incorporated in the model.

$\hat{\alpha}$: Comprises intercepts obtained during the estimation process.

$\bar{\mu}$: Constitutes a vector representing the means of factor portfolios.

$\hat{\Omega}$: Is the covariance matrix of factors, estimated as $\frac{(F - \bar{F})'(F - \bar{F})}{T-1}$, where F denotes the matrix of factors derived from excess returns in portfolios.

$\hat{\Sigma} = \frac{\hat{e}' \hat{e}}{T-L-1}$, where $\hat{e}$ is a vector of residuals.

A decreased regression intercept value indicates an increased likelihood that the GRS test will not reject the factor model⁶⁰. If the p-value of the GRS test is less than the significance level, it shows that the intercepts of models deviate from zero, and the model lacks explanatory power⁶¹.

Goodness of Fit Statistic (R^2)

The R^2 , also known as "R-squared", is regarded as a measure of how well a model fits the data. With the dependent variable positioned on the left-hand side and the independent (explanatory) variables on the right-hand side, the extent to which the regression model can account for the variance in the data is quantified by R^2 . R^2 falls within the range of 0 to 1, with interpretation

⁵⁷ [Bera et al. \(2020\)](#), p. 418.

⁵⁸ [Ahmed et al. \(2019\)](#), p. 1721.

⁵⁹ [Groot and Huij \(2018\)](#), p. 64.

⁶⁰ [Mosoeu and Kodongo \(2022\)](#), p. 58.

⁶¹ [Bera et al. \(2020\)](#), p. 422.

in terms of a percentage. A high R^2 value, close to 1, signifies that most of the variance in the data is captured by the model, whereas a low R^2 value indicates that the data is poorly accommodated by the model, explaining only a limited portion of the variance.

The goodness of fit statistic is given by:

$$R^2 = \frac{ESS}{TSS} = \frac{TSS-RSS}{TSS} = 1 - \frac{RSS}{TSS}, \quad (4.8)$$

where:

$TSS = ESS + RSS$, TSS = Total sum of squares, ESS = Explained sum of squares, RSS = Residual sum of squares

In the estimation process, I will use OLS for time-series data for each portfolio and Pooled Ordinary Least Squares (POLS) for panel data. Standard statistical and econometrical tests are implemented to test the reliability of models. To evaluate the linear regression equation, I require certain statistical metrics. One commonly used method is the t-test, which assesses hypotheses related to each specific regression slope coefficient⁶². Additionally, I will employ a t-value to assess the significance of the intercept term. The t-test is a sensible choice because it considers the standard deviation of the estimated coefficients. To examine whether the value-weighted average returns of the portfolios are other than zero, I use t-tests, too. A critical t-value is the threshold that helps me determine whether to accept or reject a hypothesis.

The t-value of the k^{th} coefficient can be computed by the formula:

$$t_k = \frac{\hat{\beta}_k}{SE(\hat{\beta}_k)} \quad (k = 1, 2, 3, \dots, k), \quad (4.9)$$

where:

The null hypothesis is: $H_0: \beta = 0$, and the alternative hypothesis is $H_1: \beta \neq 0$. I use this hypothesis for the intercept and each factor.

$\hat{\beta}_k$: Is the estimated regression coefficient of the k^{th} independent variable.

$SE(\hat{\beta}_k)$: Is the estimated standard error (the square root of the estimated variance of the distribution) of $\hat{\beta}_k$.

4.4.3. Diagnostic Tests

Before proceeding with any additional testing for the F&F factor model, it's essential to verify if the time-series and panel data regressions meet the necessary assumptions. Specifically, I

⁶² [Studenmund \(2014\), p. 128.](#)

need to assess whether the error terms exhibit signs of heteroskedasticity, whether there is a serial correlation, and whether the independent variables suffer from multicollinearity. Should any of these assumptions be violated, it could introduce bias into the estimates, affecting coefficients and standard errors. Consequently, it's crucial to conduct tests to evaluate these assumptions. The following section will outline the methods used to identify these potential characteristics. If I find heteroskedasticity or autocorrelation in my panel regressions, I use the Feasible Generalized Least Squares (FGLS) model, which solves these two problems⁶³, then we can rely on the coefficients and standard errors⁶⁴. Diagnostic tests are as follows:

1- The Variance Inflation Factor (VIF) (Multicollinearity Test)

One way to identify severe multicollinearity is to examine the variance inflation factors (VIFs) among the independent variables. This involves assessing how much one independent variable can be explained by all the other independent variables in the regression equation. The purpose of VIF is to determine whether multicollinearity is significant enough to inflate the estimated variance of a coefficient (and its standard error). High VIF values for estimated coefficients suggest a higher likelihood of substantial multicollinearity among these explanatory variables. According to [Gomez et al. \(2020\)](#)⁶⁵, low or moderate multicollinearity exists if the VIF is less than 10, which is acceptable for the regressions.

$$VIF = \frac{1}{(1-R_i^2)}. \quad (4.10)$$

2- The Durbin-Watson d Test (Autocorrelation Test)

Autocorrelation indicates that error terms exhibit correlations with each other over different observations⁶⁶. When autocorrelation is present in a series, it leads to incorrect standard error calculations, which means that while the estimates remain unbiased, they lose efficiency. As a result, T-tests and F-tests become unreliable. Moreover, the R-squared tends to be overstated in the presence of autocorrelation.

The Durbin-Watson d statistic is used to check for the presence of first-order serial correlation in the residuals⁶⁷, which are derived from the regression analysis. In this case, the hypothesis tests I conduct is one-sided:

⁶³ [Williams \(2016\)](#), p. 21.

⁶⁴ [Gullo and Montalbano \(2022\)](#), p. 15, [Sun et al. \(2021\)](#), p. 3, [Gavazzoni and Santacreu \(2020\)](#), p. 353.

⁶⁵ [Gomez et al. \(2020\)](#), p. 5.

⁶⁶ [Brooks \(2019\)](#), p. 139.

⁶⁷ [King \(1981\)](#), p. 51.

$H_0: \rho \leq 0$ (no positive serial correlation)

$H_1: \rho > 0$ (positive serial correlation),

where:

If $d < d_L$ Reject H_0

If $d > d_U$ Do not reject H_0 ; we have a negative serial correlation.

If $d_L \leq d \leq d_U$ Inconclusive.

The formula for the d-value is:

$$d = \frac{\sum_2^T (e_t - e_{t-1})^2}{\sum_1^T e_t^2}. \quad (4.11)$$

3- Breusch-Pagan (Heteroskedasticity Test)

According to the Ordinary Least Squares assumption, the error terms should have consistent variance⁶⁸, a condition known as homoscedasticity. If the error terms exhibit heteroskedasticity (varying variance), it won't affect the unbiasedness of the OLS estimated coefficients but could introduce bias in their standard errors. This, in turn, can result in unreliable hypothesis tests. Consequently, this affects the validity of statistical tests like the t-test and F-test. In my thesis, I use Breusch-Pagan with the hypotheses below:

H_0 : No heteroskedasticity

H_1 : Heteroskedasticity.

If the p-value is greater than the significance level, the null hypothesis cannot be rejected, and it will be concluded that homoscedasticity is present, and thus, t-test results can be trusted.

4- Unit root (stationarity) Test

This test is done for dependent and independent variables. When working with time series data, it's crucial to investigate whether there's a unit root in the data series. If a variable isn't stationary (meaning its statistical properties change over time), it can lead to misleadingly high R-squared values, even when there isn't a meaningful relationship between the variables. Non-stationary data can result in what's known as a "spurious regression"⁶⁹, where unrelated variables appear to be related. To avoid this issue, it's essential to check for a unit root and ensure that the data

⁶⁸ [Brooks \(2019\), p. 132.](#)

⁶⁹ [Brooks \(2019\), p. 319.](#)

is stationary before using it in linear regressions. I use the Augmented Dickey-Fuller (ADF) test as [Gonzalez and Jareno \(2019\)](#), [Chatzitzisi \(2021\)](#) did in their works.

The hypotheses are as follows:

$$\begin{array}{ll} H_0: \rho = 1 \text{ Unit root} & [\text{Variable is not stationary}] \\ H_1: \rho < 1 \text{ No unit root} & [\text{Variable is stationary}] \end{array}$$

If a coefficient significantly differs from one (specifically, if it is less than one), the hypothesis suggesting the presence of a unit root is rejected. This rejection implies stationarity in the series. Conversely, if the null hypothesis cannot be rejected, it is concluded that a unit root exists. When the p-value for a variable exceeds the significance level, it signifies the inability to reject the null hypothesis, indicating that the variable is non-stationary⁷⁰.

5- Normality test

I use the normality test to examine whether the residuals conform to a normal distribution. To do this, I apply the [Jarque-Bera test \(1987\)](#) according to [Gonzalez and Jareno \(2019\)](#), [Chatzitzisi et al. \(2021\)](#), [Horvath and Wang \(2021\)](#), [Konstantinov and Fabozzi \(2021\)](#), [Curto and Serrasqueiro \(2022\)](#), and [Bera et al. \(2020\)](#) to assess whether the data supports the following null hypothesis compared to the alternative hypothesis:

$$\begin{array}{l} H_0: \text{Residuals are normally distributed.} \\ H_1: \text{Residuals are not normally distributed.} \end{array}$$

I depended on the p-value to guide my decisions on whether to reject the null hypothesis at the significance level. If the p-value was lower than the selected significance level, I rejected the null hypothesis and supported the alternative hypothesis at that specific significance level⁷¹. The Jarque-Bera test statistic is mathematically represented as follows⁷²:

$$JB = \frac{n}{6} (S^2 + \frac{1}{4}(K - 3)^2). \quad (4.12)$$

⁷⁰ [Mohamadi and Lattanzi \(2019\)](#), p. 9.

⁷¹ [Poitras \(2006\)](#), p. 307.

⁷² [Poitras \(2006\)](#), p. 305.

Where JB referred to the Jarque-Bera test statistic, n is the number of observations, S is the sample skewness, K is the sample kurtosis. The Jarque-Bera test statistics follow a chi-squared distribution that contains two degrees of freedom.

5. Research Results and Findings

The performance of the asset pricing models, and the descriptive analysis are evaluated in this section.

[Table 1](#) shows that the six portfolios have, on average, an almost equal number of firms over 20 years. Still, as we can see, the big portfolios have the largest fraction of value (market capitalization). The average market capitalization decreases from low to high BE/ME in each size quantile. Small portfolios have, on average, greater BE/ME than large portfolios. One reason could be that the market capitalization (market valuation), which is in the denominator, is lower for small stocks compared with big stocks. This ratio proves that the market mostly undervalues small firms⁷³. Two other reasons could be information asymmetry and financial constraints⁷⁴. The lack of disclosure of sufficient information by small companies causes uncertainty among investors, and this causes investors to turn to large companies automatically. Additionally, larger firms have more bargaining power⁷⁵ to fund their projects.

	The average annual market capitalization				The average annual BE/ME		
	low	medium	high		low	medium	high
small	1,840	1,570	1,240	small	0.20	0.45	1.34
big	49,500	36,400	23,900	big	0.11	0.29	0.71
	The average annual number of firms						
	low	medium	high				
small	776.60	762.50	772.93				
big	733.59	720.53	781.84				

Table 1 Descriptive statistics for 6 stock portfolios formed on size and book-to-market equity from 2003 to 2022. The average annual market capitalization is in million U.S. dollars

⁷³ [Fontaine and Zhao \(2021\), p. 10.](#)

⁷⁴ [Fontaine and Zhao \(2021\), p. 14.](#)

⁷⁵ [Fontaine and Zhao \(2021\), p. 14.](#)

Year	Total number of firms	Year	Total number of firms	Year	Total number of firms	Year	Total number of firms
2003	662	2008	773	2013	845	2018	941
2004	666	2009	746	2014	866	2019	961
2005	688	2010	765	2015	867	2020	965
2006	717	2011	808	2016	894	2021	952
2007	744	2012	816	2017	921	2022	1,001

Table 2 Total number of firms from 2003 to 2022

[Table 2](#) shows how many companies remain each year after data processing, for instance, removing the financials and utilities, removing the companies with negative BE, etc. The number of eligible companies increases constantly from 2003 to 2022.

5.1. First step: Results of the Monthly Returns from July 2003 to December 2022

This section examines monthly data from July 2003 to December 2022.

5.1.1. Descriptive Statistics

The correlations between my factors and those from Kenneth R. French's website show how close my variables are to Fama and French. As it is obvious from [Table 1, Appendix D](#), there are strong correlations between my factors and those from the Kenneth R. French website. The correlation of SMB, HML, and market risk premium factors are 79.47%, 80.34%, and 99.60%. The reason for these high correlations is that this study was conducted in the U.S. market same as Fama and French did, and I followed the same method for factor calculation.

[Figure 1](#) shows the fluctuation of the excess return of each portfolio from July 2003 to December 2022. All the portfolios have the same trend, and the spikes and the dips take place simultaneously in this period. The small-high portfolio has the highest fluctuations among all six portfolios. Small stocks have less liquidity. Therefore, the low level of liquidity makes small stocks more sensitive to market changes and experience larger price fluctuations with the increase of buyers and sellers⁷⁶. In addition, stocks with a higher book-to-market ratio may have higher returns because investors think that the market price of these stocks is lower than their book value, and in the future, their prices will increase and they will buy them, which leads to

⁷⁶ [Lof and Bommel \(2023\), p. 9.](#)

an increase in demand for stocks. Finally, we observe the increase in stock prices and returns. Another reason was explained in capital 3.1.2. with equation 3.4.

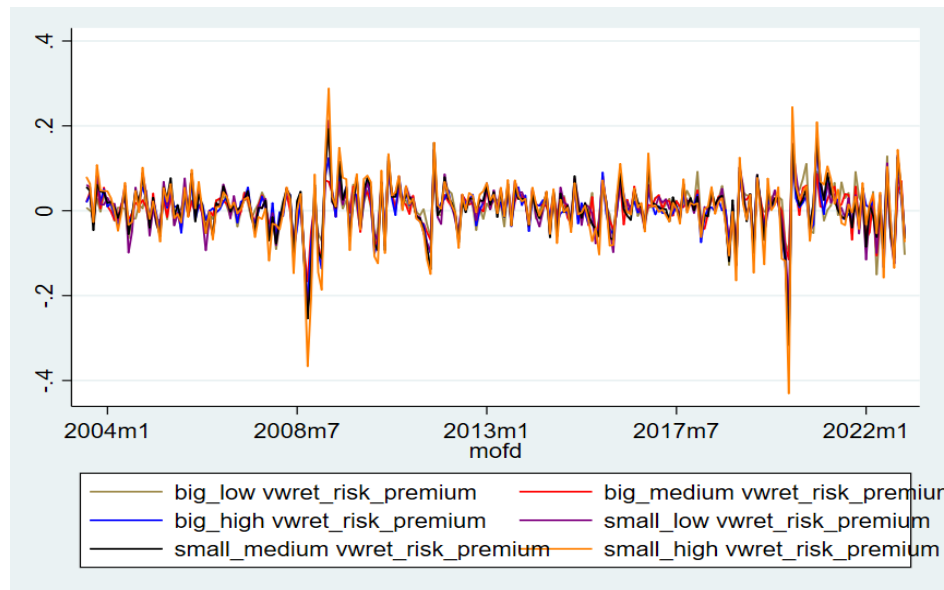


Figure 1 Fluctuation of the monthly portfolio excess returns. In this study, `vwret_risk_premium` stands for portfolio value-weighted average returns minus the risk-free rate

Figure 1 also illustrates that the portfolios have major spikes and drops in 2008-2009 and 2020-2021. As [Horvath and Wang \(2021\)](#) proved that significant events like financial events and economic conditions affect the stock market, we can see in the graph that the well-known financial crisis in 2008-2009 and the COVID-19 pandemic in 2020-2021 influenced the fluctuation of the stock prices, and the portfolios record the highest and lowest returns during these two events. [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#) and [Galsband \(2012\)](#) mention in their studies that during certain events, such as 2008 and 2020, the returns of low BE/ME stocks and high BE/ME stocks have the largest gap with each other. As [Figure 1](#) indicates, the stocks with high BE/ME are more sensitive to downside risk than stocks with low BE/ME.

[Table 3](#) shows the descriptive statistics (Mean and standard deviation) of each portfolio's value-weighted average return from July 2003 to December 2022. As it is known, when we go from the side of small-sized to large-sized portfolios, the average return of the portfolios decreases. So, we can assume that the size has an inverse relationship with the average portfolio return. This is a confirmation of Fama and French studies in [1993](#) and [2015](#), [Groot and Huij \(2018\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), and [Mosoeu and Kodongo \(2022\)](#). One of the reasons why small portfolios or generally small stocks have higher average returns than big portfolios is that they have higher fluctuations (standard deviations)⁷⁷ than big stocks, so they

⁷⁷ [Bera et al. \(2020\), p. 417.](#)

could reach higher peaks. Small firms have a higher cost of capital than larger companies⁷⁸; in other words, they are riskier, and because of their high-risk tolerance, compared to large companies, they have more returns⁷⁹. This result is consistent with [Bera et al. \(2020\)](#). In small portfolios, we can see that the portfolios with higher BE/ME have higher portfolio returns, but in big portfolios, we can see that the average return decreases from low to high book-to-market ratio, which confirms the results of [Feyyaz et al. \(2019\)](#) and [Mosoeu and Kodongo \(2022\)](#). So, the BE/ME negatively relates to average return in big portfolios and positively in small portfolios. Furthermore, big-high portfolio has a lower average return than small-low portfolio, as [Mosoeu and Kodongo \(2022\)](#) concluded in their study.

portfolio	Summary of value-weighted average return of each portfolio		
	Mean	Std. dev.	Freq.
big high	.00442312	.04317942	234
big low	.00628555	.04124673	234
big medium	.00573318	.03796966	234
small high	.00727111	.06035309	234
small low	.00719809	.04969353	234
small med..	.00751349	.049616	234
Total	.00640409	.04750394	1,404

Table 3 Descriptive statistics of portfolios' value-weighted average returns (monthly data). The values of mean and standard deviation are not in percentage. To convert the values to percentages, multiply them by 100

The t-tests (see [Appendix D, Table 2](#)) show that the mean value-weighted average returns of small-low, small-medium, big-low, and big-medium portfolios with t-statistics of 2.21, 2.31, 2.33, and 2.31 significantly differentiate from zero, but testing the mean value-weighted average returns of small-high and big-high portfolios, which is the lowest return among all portfolios, with t-statistics of 1.84 and 1.57 illustrates that we cannot refuse the null hypothesis that is the average return is equal zero. The small-medium portfolio has the highest average return. After that, small-high has the highest average return; finally, one of the hypotheses of this study, which is that firms with smaller sizes and higher book-to-market ratios can generate higher average returns among other portfolios, has been proved. The average return of all portfolios is positive, and small portfolios have higher volatilities ([Bera et al. \(2020\)](#)).

[Table 4](#) shows the descriptive statistics of the three factors. The mean of market risk and SMB factors are positive. However, this value is for the HML factor negative. This means that, on

⁷⁸ [Yang et al. \(2023\)](#), p. 7.

⁷⁹ [Fama and French \(1996\)](#).

average, small stocks (portfolios) outperform big stocks (portfolios) over the given period. Additionally, portfolios with low book-to-market ratios have higher returns than portfolios with high book-to-market ratios, which [Bera et al. \(2020\)](#) concluded the same. Value stocks are riskier than growth stocks⁸⁰, and investors who buy this type of stock believe that the stock still has the potential to grow. The average market risk premium is 0.51%, which means that, on average, the market return is 0.51% above the risk-free rate. The market risk premium has the highest volatility and average return among these three factors.

Variable	Obs	Mean	Std. dev.	Min	Max
Market_Risk	234	.0051086	.0383527	-.0844984	.0661516
SMB	234	.0020485	.0244264	-.0427172	.0459362
HML	234	-.0009901	.0243244	-.0440673	.0462173

Table 4 Descriptive statistics of risk factors (monthly data). The mean, standard deviation, minimum, and maximum values are not in percentage. To convert the values to percentages, multiply them by 100

5.1.2. Time-series Regressions

After introducing and analyzing the variables statistically, I used the time-series OLS regression model introduced in equation 3.2 to understand how significantly the SMB, HML, and market risk premium can explain the variations of each portfolio's excess return. [Table 5](#) contains the coefficients of the variables with the t-statistics in parenthesis and the adjusted R-squared of each portfolio, indicating the explanatory power of the risk factors. The regression outputs are available in [Appendix D, Table 3](#).

The adjusted R-squared of the portfolios ranges from 93.13% to 96.38% and is very close to the outputs of [Bera et al. \(2020\)](#) and Kenneth R. French. The small-high portfolio has the highest explanatory power, unlike the big-medium with the lowest explanatory power. The coefficients of all factors for all portfolios are statistically significant (the significance level in this study is 5%) except for SMB in the big-high portfolio. [Bera et al. \(2020\)](#) mention that getting an insignificant SMB coefficient in the Fama and French models is common. Overvalued firms with high book-to-market ratios and substantial market capitalizations are included in big-high portfolio. Thus, it is probable that overpriced companies' returns are unaffected by SMB⁸¹. Additionally, HML in small-medium and big-medium portfolios is insignificant. The market risk factor has the most significant coefficient. It has the highest correlations in all six portfolios' excess returns ([Bera et al. \(2020\)](#)), and the relation between the market factor and

⁸⁰ [Kang et al. \(2011\), p. 3158.](#)

⁸¹ [Bera et al. \(2020\), p. 419.](#)

the average excess return of the portfolios is positive. It means the market risk premium captures the changes in stock return more than SMB and HML. This finding is consistent with Fama and French (1993, 2015, 2016) and Bera et al. (2020). As discussed in the descriptive analysis section, we expect that SMB has a negative correlation with the excess return of big portfolios and a positive correlation with small portfolios (Fama and French 1993). In addition, the HML factor should have a negative correlation with the average excess returns of lower book-to-market ratio portfolios and a positive correlation with the average excess returns of high book-to-market ratios (Fama and French 1993). The results of this study in Table 5 support these hypotheses. In each size group, the coefficient of the HML increases from low to high book-to-market ratio portfolios. In each value group, the coefficient of SMB decreases from small to big portfolios due to the method of the factors' calculation and portfolios' construction⁸².

Coefficient (SMB)				Coefficient (HML)			
	low	medium	high		low	medium	high
small	0.97 (30.20)	0.85 (25.89)	0.90 (23.42)	small	-0.43 (-14.24)	-0.01 (-0.40)	0.48 (13.24)
big	-0.08 (-2.51)	-0.12 (-3.53)	-0.00 (-0.15)	big	-0.41 (-12.96)	-0.00 (-0.14)	0.56 (16.71)
Coefficient (market risk premium)				Adj. R-squared			
	low	medium	high		low	medium	high
small	0.98 (54.26)	0.94 (51.23)	1.06 (49.21)	small	96.23%	96.06%	96.38%
big	1.09 (57.21)	0.99 (52.72)	0.95 (47.30)	big	93.92%	93.13%	93.88%

Table 5 Summary results of time-series regressions for each portfolio (monthly data)

5.1.3. Compare with Kenneth R. French

To determine how close my results are to the Fama-French studies, I ran the same regressions in the previous section for the collected data from the Kenneth R. French database. The results can be seen in Table 6. We can see that the results of this table are very close to the results of extracted data from S&P Composite 1500. The R-squared ranges from 89.17% to 95.35%. The

⁸² Bera et al. (2020), p. 419.

coefficient of the SMB factor in big-medium and big-high portfolios is insignificant. SMB in small portfolios has a positive relation (coefficient) with the portfolio excess returns, and in the big portfolios, a negative relation (coefficient). Furthermore, the HML factor in portfolios with higher BE/ME has a higher coefficient. The coefficients of the market factor show the highest correlation with the average excess return. You find the outputs of the regressions of this section in [Appendix D, Table 6](#).

The few differences between the results of this thesis in [Table 5](#) and the Kenneth R. French database in [Table 6](#) are because I have a different reference market from that the Fama and French use in their studies. They use all the companies on NASDAQ, AMEX, and NYSE, but I use the S&P Composite 1500, which contains only 1500 companies from NASDAQ and NYSE.

	Coefficient (SMB)				Coefficient (HML)		
	low	medium	high		low	medium	high
small	1.24 (25.59)	0.94 (25.31)	1.01 (23.17)	small	-0.26 (-6.25)	0.35 (10.78)	0.87 (22.98)
big	-0.08 (-2.87)	-0.00 (0.03)	0.09 (1.53)	big	-0.29 (-11.56)	0.30 (7.91)	0.94 (18.21)
	Coefficient (market risk premium)				Adj. R-squared		
	low	medium	high		low	medium	high
small	1.15 (41.35)	1.08 (50.57)	1.11 (44.22)	small	93.50%	95.35%	94.94%
big	1.07 (63.93)	1.04 (41.66)	1.20 (35.22)	big	95.00%	89.95%	89.17%

Table 6 Summary results of time-series regressions for each portfolio for Kenneth R. French data (monthly data)

5.1.4. GRS Test

As mentioned in Section 4.4.2, if the p-value of the GRS test is above the significance level, we cannot conclude that the intercepts of regression models are jointly and significantly different from zero; therefore, we cannot reject the null hypothesis. The results of the GRS test in [Appendix D, Table 4](#) are prepared for the one-factor model (only market risk premium), two-factor models (market risk premium and SMB, market risk premium and HML), and the three-factor model (market risk premium, SMB, and HML). The p-values of the GRS tests are

subsequently 0.62, 0.58, 0.74, and 0.65, which means that the GRS test cannot reject that the alphas in the three-factor model are jointly different from zero. Therefore, we cannot reject the null hypothesis. This means that the three factors significantly capture all the variations in average return. This is a confirmation of [Fama and French \(1993\)](#), [Liu et al. \(2019\)](#), [Bera et al. \(2020\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), and [Horvath and Wang \(2021\)](#) studies. As we can see from [Appendix D, Table 4](#), the adjusted R-squared of the tests increases by adding the factors. So, the three-factor model is the best model among the four models in effectively explaining the excess returns of portfolios.

5.1.5. Regression for Panel Data

The next step is to run the regression for the panel data for the whole period to test whether our three risk factors can explain the variations in all six portfolios' excess returns together. Panel data analysis has three models: 1- Pooled OLS Model (POLS), 2- Fixed Effect Model (FE), and 3- Random Effect Model (RE)⁸³. To determine which model is suitable for data analysis and regression; it is necessary to perform the Breusch and Pagan Lagrangian multiplier test⁸⁴. The null hypothesis in this test is no panel effect, and if the p-value is above the significance level, then POLS is preferred. If the p-value is less than the significance level, then I will run the Hausman test to decide between a fixed or a random effect model⁸⁵. In this study, the p-value of this test is 1.00 (see [Appendix D, Table 7](#)), which is much more than 5%, therefore, the POLS would suffice, and we don't need FE and RE anymore.

In [Appendix D, Table 8](#), you can see the regression output of the monthly data. The overall R-squared of the regression is 0.7958, which means that the SMB, HML, and market risk premium can jointly explain 79.58% of the variations in portfolios' excess returns, which is a good result. The coefficient of market risk premium, SMB, and HML are respectively 1.16, 0.50, and 0.13, and they are all statistically significant, which is consistent with the results of [Fama and French \(1993\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), [Feyyaz et al. \(2019\)](#), [Hanauer et al. \(2024\)](#), [Liu et al. \(2019\)](#). These factors have positive relations with the changes in portfolios' excess returns. The intercept of the regression model is significant and very close to zero. The intercept shows that only a small part of the volatility in portfolio excess return is not explained by the independent variables, and the model has no abnormal return.

⁸³ [Brooks \(2019\)](#), p 490-501.

⁸⁴ [Mia \(2022\)](#), p. 2, [Melo and Graham \(2014\)](#), p. 43, [Sequeira and Morao \(2020\)](#), p. 73.

⁸⁵ [Peng et al. \(2017\)](#), p. 5.

In the next section, it will be explained that this regression suffers from heteroskedasticity. Therefore, the FGLS model was used to solve this problem. The output of this regression can be seen in [Appendix D, Table 9](#).

5.1.6. Diagnostic Tests for Time-series and Panel Data Regressions

In this part, I interpret the results of diagnostic tests for each time-series regression, then for POLS:

I start with the stationary test (ADF test) for each variable. As you can see from the results of the ADF test in [Appendix D, Table 10](#), all the variables in this study are stationary because the absolute value of the test statistic from the ADF test is greater than the absolute value of the critical value. So, there is no problem running the regressions. Furthermore, The VIF for each variable is far less than 10 (see [Appendix D, Table 11](#)). Therefore, there is no multicollinearity among the independent variables.

OLS regression portfolio small-low:

The p-value of the Breusch-Pagan test is 0.19, which is above 0.05. Therefore, we cannot reject the null hypothesis, which stands for no heteroskedasticity. The boundaries of the Durbin-Watson test (d_L and d_U) are respectively 1.75 and 1.80⁸⁶. The Durbin-Watson statistic is 2.06 which is greater than the upper critical value, therefore, there is a negative autocorrelation in the model. The p-value of the JB test is 0.22 which is above the significance level, so the regression residuals are normally distributed.

OLS regression portfolio small-medium:

The p-value of the Breusch-Pagan test is 0.17 which is above 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The Durbin-Watson statistic is 2.01 which is greater than the upper critical value, therefore, there is a negative autocorrelation in the model. The p-value of the JB test is 0.66, which is above the significance level, so the regression residuals are normally distributed.

OLS regression portfolio small-high:

The p-value of the Breusch-Pagan test is 0.16, which is above 0.05. Therefore, we cannot reject the null hypothesis, which stands for no heteroskedasticity. The Durbin-Watson statistic is 2.29, which is greater than the upper critical value. Therefore, there is a negative autocorrelation in

⁸⁶ See <https://real-statistics.com/statistics-tables/durbin-watson-table/> [last accessed on 23.02.2024].

the model. The p-value of the JB test is less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio big-low:

The p-value of the Breusch-Pagan test is 0.30, which is above 0.05. Therefore, we cannot reject the null hypothesis, which stands for no heteroskedasticity. The Durbin-Watson statistic is 2.12, which is greater than the upper critical value. Therefore, there is a negative autocorrelation in the model. The p-value of the JB test is 0.16, which is above the significance level, so the regression residuals are normally distributed.

OLS regression portfolio big-medium:

The p-value of the Breusch-Pagan test is 0.19, which is above 0.05. Therefore, we cannot reject the null hypothesis, which stands for no heteroskedasticity. The Durbin-Watson statistic is 2.17, which is greater than the upper critical value. Therefore, there is a negative autocorrelation in the model. The p-value of the JB test is 0.16, which is above the significance level, so the regression residuals are normally distributed.

OLS regression portfolio big-high:

The p-value of the Breusch-Pagan test is 0.01, which is less than 0.05 and shows heteroskedasticity in the model. The Durbin-Watson statistic is 2.13 which is greater than the upper critical value, therefore, there is a negative autocorrelation in the model. The p-value of the JB test is less than the significance level, so the regression residuals are not normally distributed.

POLS regression for panel data:

Some diagnostic tests for panel data are different than ones for time-series data, so to obtain more accurate results, tests that are appropriate to the panel data should be applied in this section. The Levin-Lin-Chu unit root test⁸⁷ shows that all the variables in this study are stationary, so we can run the POLS regression. In this test, the null hypothesis indicates that panels contain unit roots, and the alternative hypothesis shows that the panels are stationary. We can reject the null hypothesis if the p-value is less than the significance level. You can see the results of this test in [Appendix D, Table 30](#). The VIF test shows no multicollinearity in the model because the VIF is less than 10 for each variable. For testing the autocorrelation in my

⁸⁷ There are plenty of studies that used this test, such as [Ahmad et al. \(2022\)](#) and [Szyszko \(2022\)](#).

panel data, I use the Wooldridge test as [Fier et al. \(2013\)](#) did in their study. In this test, the null hypothesis is no panel autocorrelation among the residuals. We cannot reject the null hypothesis if the p-value is above the significance level. In this study, the p-value of the test is 0.38, which is well above 0.05, so we have no autocorrelation in the panel data. The Breusch Pagan test shows that we have heteroskedasticity. To tackle this problem, I use the FGLS model so that we can rely on the coefficients and the p-values. It is to be mentioned that in this method, the standard errors will be corrected without changing the coefficients and the R-squared. The Jarque Bera test shows that the residuals are not normally distributed.

5.2. Second step: Results of the Daily Returns from July 2018 to December 2022

This section examines daily data from July 2018 to December 2022. Since many of the results obtained from daily data are the same as those obtained from monthly data, additional interpretations and reasonings have been avoided in some of the following parts to prevent repetition.

5.2.1. Descriptive Statistics

The daily risk factors, just like the monthly data show high correlations with the daily factors extracted from the Kenneth R. French database. As it is obvious from [Table 1 in Appendix E](#), the correlations of SMB, HML, and market risk premium factors are 78.69%, 83.66%, and 99.62%.

[Figure 2](#) shows the fluctuation of the excess return of each portfolio from July 2018 to December 2022. All portfolios have the same trend, and the spikes and the dips take place simultaneously. The small-high portfolio has the highest fluctuations among all six portfolios. The graph also illustrates that the portfolios have major spikes and drops (volatility) during the COVID-19 pandemic in 2020.

[Table 7](#) shows the descriptive statistics (Mean and standard deviation) of each portfolio's value-weighted average return from July 2018 to December 2022. As it is known, when we go from the side of small-sized portfolios to large-sized portfolios, the average return of the portfolios decreases. As a result, the size factor has an inverse relationship with the average return. This is a confirmation of Fama and French studies in [1993](#) and [2015](#), [Groot and Huij \(2018\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), and [Mosoeu and Kodongo \(2022\)](#). In small portfolios, by increasing the BE/ME, the average portfolio return increases, but in big portfolios, we can see that the average portfolio return decreases from a low to a high book-to-market ratio, which confirms the results of [Feyyaz et al. \(2019\)](#) and [Mosoeu and Kodongo \(2022\)](#). So, the value

factor has a negative relation with the average return in big portfolios and a positive relation in small portfolios.

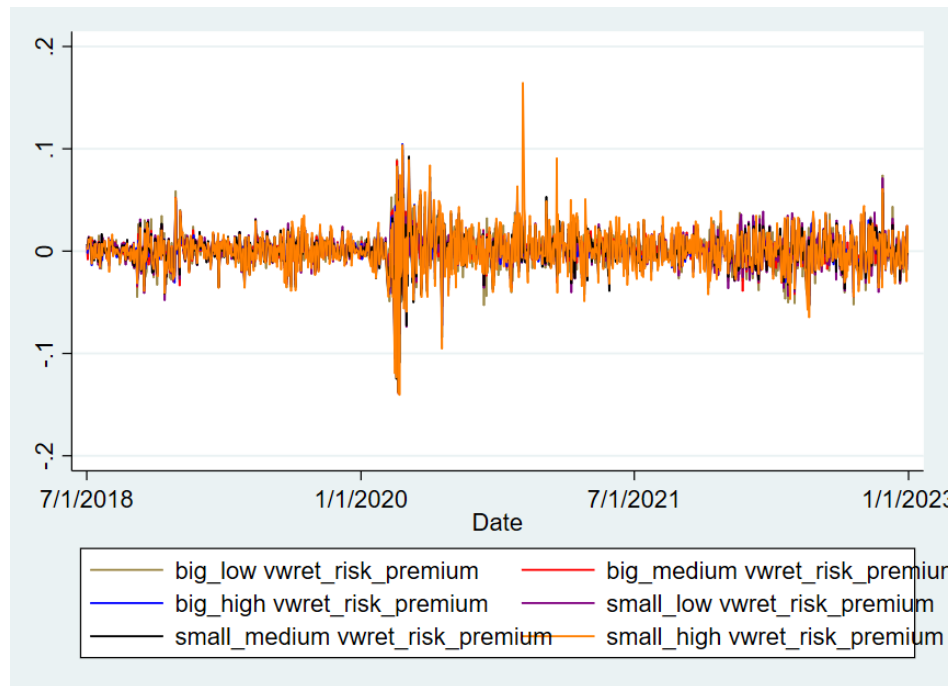


Figure 2 Fluctuation of the daily portfolio excess returns. In this study, `vwret_risk_premium` stands for portfolio value-weighted average returns minus the risk-free rate

portfolio	Summary of value-weighted average return of each portfolio		
	Mean	Std. dev.	Freq.
big high	.00049383	.01102599	1,134
big low	.00070139	.01282643	1,134
big medium	.0005204	.01012014	1,134
small high	.00118132	.0161701	1,134
small low	.00083891	.01374667	1,134
small med..	.00093462	.01362348	1,134
Total	.00077841	.01306484	6,804

Table 7 Descriptive statistics of portfolios' value-weighted average returns (daily data). The values of mean and standard deviation are not in percentage. To convert the values to percentages, multiply them by 100

The t-tests (see [Appendix E, Table 2](#)) show that the mean value-weighted average return of small-low, small-medium, and small-high portfolios with t-statistics of 2.05, 2.31, and 2.46 significantly differentiate from zero, but testing the mean value-weighted average returns of big-low, big-medium, and big-high portfolios with t-statistics of 1.84, 1.73, and 1.51 illustrates that we cannot refuse the null hypothesis that is the average return is equal zero. The small-high

portfolio has the highest average return; after that, small-medium has the highest average return. In the previous part related to the monthly data from July 2003 to December 2022, we observed that these two portfolios also had the highest value-weighted average returns too. Finally, one hypothesis of this study, which is that firms with smaller sizes and higher book-to-market ratios can generate higher average returns among other portfolios, has been proved.

[Table 7](#) and [Table 8](#) indicate that all series' means are positive; this indicates that the investors will benefit by taking on the risk⁸⁸. This means that, on average, small stocks (portfolios) outperform big stocks (portfolios). Additionally, portfolios with higher book-to-market ratios have higher returns than portfolios with low book-to-market ratios. According to Fama and French (2007), value stocks (low BE/ME) could be subject to extraordinary financial pressure that only appears in the most extreme financial crises. The average market risk premium is 0.03%, which means that on average the market return of the S&P Composite 1500 is 0.03% above the risk-free rate. Among these three factors, the market risk premium has the highest volatility and average return.

Variable	Obs	Mean	Std. dev.	Min	Max
Market_Risk	1,134	.0003552	.0102215	-.0219761	.0189833
SMB	1,134	.0002246	.0067985	-.0125003	.0131595
HML	1,134	.0000214	.008017	-.0149415	.0159579

Table 8 Descriptive statistics of risk factors (daily data). The mean, standard deviation, minimum, and maximum values are not in percentage. To convert the values to percentages, multiply them by 100

5.2.2. Time-series Regressions

[Table 9](#) contains the coefficient of the variables with the t-statistics in parenthesis as well as the adjusted R-squared of each portfolio, which is an indicator of the explanatory power of the risk factors. The regression outputs are available in [Appendix E, Table 3](#).

The adjusted R-squared of the portfolios ranges from 93.01% to 95.43% and is very close to the outputs of [Bera et al. \(2020\)](#) and Kenneth R. French. Unlike the monthly data, where the small-high portfolio had the most explanatory power, in this section, the daily factors of the small-low portfolio have the highest explanatory power. In general, it can be concluded from the monthly and daily data outputs that smaller portfolios have higher explanatory power. The coefficients of all factors for all portfolios are statistically significant except for SMB in the big-high portfolio and HML in the small-medium portfolio. The market risk factor has the most significant and highest coefficients in all six portfolios, and the relation between the market

⁸⁸ [Bera et al. \(2020\), p. 417.](#)

factor and the excess return of the portfolios is positive. So, it means that the market risk premium captures the changes in stock return more than SMB and HML. This finding is consistent with Fama and French (1993, 2015, 2017) and Bera et al. (2020). As discussed in the descriptive analysis section, we expect that SMB has a negative correlation with the excess return of big portfolios and a positive correlation with small portfolios. In addition, the HML factor should have a negative correlation with lower book-to-market ratio portfolios and a positive correlation with high book-to-market ratios. The results of this study in Table 9 support these hypotheses. In each size group, the coefficient of the HML increases from low to high book-to-market ratio portfolios, and in each value group, the coefficient of SMB decreases from small to big portfolios.

	Coefficient (SMB)				Coefficient (HML)		
	low	medium	high		low	medium	high
small	0.95 (59.15)	0.91 (51.87)	0.94 (40.08)	small	-0.34 (-25.48)	-0.014 (-1.00)	0.53 (27.33)
big	-0.05 (-3.26)	-0.089 (-6.66)	0.03 (1.87)	big	-0.42 (-30.50)	-0.04 (-3.72)	0.55 (41.76)
	Coefficient (market risk premium)				Adj. R-squared		
	low	medium	high		low	medium	High
small	1.03 (113.03)	1.02 (102.67)	1.12 (84.51)	small	95.43%	94.53%	93.01%
big	1.14 (119.39)	0.97 (127.38)	0.99 (110.56)	big	94.32%	94.19%	93.21%

Table 9 Summary results of time-series regressions for each portfolio (daily data)

5.2.3. Compare with Kenneth R. French

We can see that the results of Table 10 are very close to the results of extracted data from the S&P Composite 1500. The R-squared ranges from 90.75% to 98.17%. The coefficients of all factors in all portfolios are statistically significant. The SMB factor has a negative correlation with the excess return of portfolios, which means that smaller portfolios have higher coefficients than big portfolios. Furthermore, the HML factor has a positive relation with the excess return which means in every size quantile the portfolios with higher book-to-market ratios have higher and positive coefficients than portfolios with lower book-to-market ratios.

The coefficients of the market factor show the highest correlation with the average excess return. To see the output of the regressions of this section, you can see [Appendix E, Table 6](#). By comparing [Table 5](#) with [Table 6](#), and [Table 9](#) with [Table 10](#), we can conclude that there are really small differences between the results from the S&P Composite 1500 and Kenneth R. French database. Because Kenneth R. French is a very reliable source and besides the fact that the results of this research so far prove the results of Fama and French in 1993, the factors and results of this master's thesis can be reliable.

	Coefficient (SMB)				Coefficient (HML)		
	low	medium	high		low	medium	high
small	1.09 (57.17)	0.85 (54.62)	0.84 (49.71)	small	-0.17 (-14.21)	0.32 (33.02)	0.70 (64.38)
big	-0.14 (-18.31)	0.03 (2.20)	0.06 (3.15)	big	-0.22 (-46.05)	0.32 (37.53)	0.75 (61.58)
	Coefficient (market risk premium)				Adj. R-squared		
	low	medium	high		low	medium	high
small	1.10 (97.89)	1.00 (109.38)	0.99 (95.58)	small	94.28%	94.60%	93.87%
big	1.04 (225.54)	0.90 (110.32)	1.04 (90.58)	big	98.17%	92.32%	90.75%

Table 10 Summary results of time-series regressions for each portfolio for Kenneth R. French data (daily data)

5.2.4. GRS Test

The results of the GRS test in [Appendix E, Table 4](#) are prepared for the one-factor model (only market risk premium), two-factor models (market risk premium and SMB, market risk premium and HML), and the three-factor model (market risk premium, SMB, and HML). As “grsftest” syntax does not work for databases with more than 800 observations in Stata, I use “grstest2” syntax, therefore, we should consider the mean absolute alpha. After adding more factors to the models, we see that the mean absolute alpha, which is an indication of intercept, reduces and it is very close to zero. This means that the three-factor model significantly captures all variations in the portfolio excess returns and performs better than CAPM which has the least power to capture the variations among these three models. This a confirmation of [Fama and French](#)

(1993), [Liu et al. \(2019\)](#), [Bera et al. \(2020\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), and [Horvath and Wang \(2021\)](#) studies.

5.2.5. Regression for Panel Data

The p-value of the Breusch and Pagan Lagrangian multiplier test is 1.00 (see [Appendix E, Table 7](#)) which is greater than 5%, therefore, the POLS would suffice, and we don't need the FE model and RE model anymore.

In [Appendix E, Table 8](#) you can see the regression output of the daily data. The overall R-squared of the regression is 0.7115, which means that the SMB, HML, and market risk premium can jointly explain 71.15% of the variation in all portfolios' excess returns, which is a good result. The coefficient of market risk premium, SMB, and HML are respectively 1.29, 0.52, and 0.13 and they are all statistically significant and have positive relation with the changes in portfolio excess returns, which is consistent with the results of [Fama and French \(1993\)](#), [Ugurlu-Yildirim and Sendeniz-Yuncu \(2021\)](#), [Feyyaz et al. \(2019\)](#), [Hanauer et al. \(2024\)](#), [Liu et al. \(2019\)](#). According to [Bera et al.'s](#) study in 2020, by increasing the time horizons the coefficient between market risk premium and average stock returns shows a stronger relationship, which is rejected in my research, because the related coefficient in daily intervals, which has the shorter period than monthly interval, is statistically more significant and has a higher correlation with the excess returns. However, this study proves the explanation of [Bera et al. \(2020\)](#) that adjusted R^2 has higher power at higher time scales (in my thesis the monthly data), and it is an indication of a strong relationship between risk and return in longer periods. In the next section, it will be explained that this regression suffers from heteroskedasticity. Therefore, the FGLS model was used to solve this problem. The output of this regression can be seen in [Appendix E, Table 9](#).

5.2.6. Diagnostic Tests for Time-series and Panel Regressions

In this part, I interpret the results of diagnostic tests for each time-series regression and after that for the panel regression:

As you can see from the results of the ADF test in [Appendix E, Table 10](#), all the variables in this study are stationary, because the absolute value of the test statistic from the ADF test is greater than the absolute value of the critical value. So, there is no problem running the regressions. Furthermore, [Table 11 in Appendix E](#) indicates that the VIF for each variable is far less than 10. Therefore, there is no multicollinearity among the independent variables.

OLS regression portfolio small-low:

The p-value of the Breusch-Pagan test is 0.11 which is above 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The boundaries of the Durbin-Watson test (d_L and d_U) are respectively 1.90 and 1.91. The Durbin-Watson statistic is 1.69 which is less than the lower critical value, therefore, there is a positive autocorrelation in the model. The p-value of the JB test is almost 0.00 which is less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio small-medium:

The p-value of the Breusch-Pagan test is 0.07 which is above 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The Durbin-Watson statistic is 1.53 which is less than the lower critical value, therefore, there is a positive autocorrelation in the model. The p-value of the JB test is almost 0.00 which is less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio small-high:

The p-value of the Breusch-Pagan test is 0.00 which is less than 0.05 and shows heteroskedasticity in the model. The Durbin-Watson statistic is 1.57 which is less than the lower critical value, therefore, there is positive autocorrelation in the model. The p-value of the JB test is 0 and less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio big-low:

The p-value of the Breusch-Pagan test is 0.39 which is above 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The Durbin-Watson statistic is 1.53 which is lower than the lower critical value, therefore, we reject the null hypothesis and there is positive autocorrelation in the model. The p-value of the JB test is almost 0.00 which is less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio big-medium:

The p-value of the Breusch-Pagan test is 0.06 which is greater than 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The Durbin-Watson statistic is 1.47 which is less than the lower critical value, therefore, we reject the null hypothesis and

there is a positive autocorrelation in the model. The p-value of the JB test is almost 0.00 which is less than the significance level, so the regression residuals are not normally distributed.

OLS regression portfolio big-high:

The p-value of the Breusch-Pagan test is 0.84 which is above 0.05, therefore, we cannot reject the null hypothesis which stands for no heteroskedasticity. The Durbin-Watson statistic is 1.67 which is less than the lower critical value, therefore, we reject the null hypothesis and there is a positive autocorrelation in the model. The p-value of the JB test is almost 0.00 and less than the significance level, so the regression residuals are not normally distributed.

POLS regression for panel data:

The Levin-Lin-Chu unit root test shows that all the variables in this study are stationary, so, we can run the POLS regression. You can see the results of this test in [Appendix E, Table 30](#). The VIF test shows that there is no multicollinearity in the model because the VIF for each independent variable is less than 10. The p-value of the Wooldridge test is 0.07 which is above 0.05, so, we have no autocorrelation in the panel data. The Breusch Pagan test shows that we have heteroskedasticity. To tackle this problem, I use the FGLS model, so that we can rely on the coefficients and the p-values. The JB test indicates that the error terms are not normally distributed.

The comparison between the results of two POLS models (monthly and daily) illustrates that both models prove the results of the Fama and French three-factor model in 1993. The R^2 of daily panel regression is 8.43% less than monthly panel regression. The factor coefficients of daily intervals are generally a little above the factor coefficients of monthly intervals and they are more significant for the daily interval. [Fama and French \(1997\)](#) state that a period of unexpectedly greater cash flows may arise if the coefficients of SMB and HML factors at long-term investment horizons are fewer than those at short-term. [Bera et al. \(2020\)](#) and [Fama and French \(1997\)](#) also concluded in their study that the coefficients and R-squared change over different time scales. The intercept of the monthly regression model is statistically significant and very close to zero, which means that there is no evidence of abnormal return in the model and the three factors can perfectly explain the fluctuations in the portfolio excess returns. On the other hand, the alpha of the daily panel regression is closer to zero than the monthly intercept but insignificant. It implies that given the portfolio's exposure to size, value, and market risk, its returns are consistent with expectations. In this instance, the absence of statistical significance suggests that there is no discernible difference between the observed intercept and

zero, and any variation from zero may be the result of random chance rather than a regular trend. So, we can take this conclusion that the three factors can perfectly explain the fluctuations in the portfolio excess returns in both monthly and daily intervals.

5.3. Third step: Results of the Daily Returns with Bitcoin from July 2018 to December 2022

In this section, I add the return of Bitcoin to the daily model for the period from July 2018 to December 2022. The number of firms is 29, which contains all the blockchain-based companies in the S&P Composite 1500.

5.3.1. Descriptive Statistics

[Table 11](#) shows the descriptive statistics of Bitcoin return. The average return of Bitcoin return is positive and very close to zero. The table also indicates that the volatility of Bitcoin return is greater than the other three factors in [Table 8](#).

Variable	Obs	Mean	Std. dev.	Min	Max
bitcoin_retn	1,134	.0003125	.0302467	-.0618378	.059904

Table 11 Descriptive statistics of Bitcoin return (daily data). The values of mean, standard deviation, minimum, and maximum are not in percentage. To convert the values to percentages, multiply them by 100

5.3.2. GRS Test

The results of the GRS test in [Appendix F, Table 2](#) are prepared for the four-factor model (logarithm of market risk premium (Lmarket risk premium), the logarithm of SMB (LSMB), the logarithm of HML (LHML), and the logarithm of Bitcoin return (Lbitcoin return)). The mean absolute alpha, which is an indication of intercept, is greater than zero and statistically significant. This means that the four-factor model cannot capture all the variations of the portfolio excess returns, and it suggests that there are abnormal returns in the model⁸⁹. Based on the GRS test, the Fama-French three-factor model works better than the four-factor model in this research.

5.3.3. Regression for Panel Data

The next step is to run the regression for the panel data containing 29 firms for the period from July 2018 to December 2022 to test the effect of adding the return of Bitcoin on the model. The

⁸⁹ [Bera et al. \(2020\), p. 418.](#)

p-value of the Breusch and Pagan Lagrangian multiplier test is above 0.05, therefore the POLS model is preferred. The output of the test is in [Appendix F, Table 1](#).

In [Appendix F, Table 3](#), you can see the regression output of the POLS model. The overall R-squared of the regression is 0.1931, which means that the factors can jointly explain 19.31% of the variations in portfolios' excess returns. As mentioned in section 3.3 the main reason for the reduction in the explanatory power is that cryptocurrencies differ from stocks in terms of their unique characteristics such as volatility. Furthermore, we have only 29 firms in this stage which is less than the sample size in steps one and two. The coefficient of Lmarket risk premium, LSMB, and Lbitcoin return are statistically significant, but the coefficient of the LHML with the value of -0.4 is statistically insignificant and appears to be redundant in the model. As it is explained in the next section, this regression model faces heteroskedasticity. To have a homoscedastic model without autocorrelation I used FGLS, which gives us reliable coefficients. [Table 3 in Appendix F](#) indicates that the coefficients of Lmarket risk premium, LSMB, and Lbitcoin return are respectively 0.37, 0.21, and 0.13 and significant after running FGLS. According to the results, Bitcoin has the role of diversifier for stock portfolios and investors can add Bitcoin to their stock portfolios to diversify their portfolios because the coefficient of Bitcoin return is positive and weak. [Gil-Alana et al. \(2020\)](#) also concluded that Bitcoin has a diversifying role, however, some of the previous researchers such as [Dyhrberg \(2016\)](#), [Huang et al. \(2021\)](#), and [Mariana \(2021\)](#) confirm that Bitcoin has a hedging and safe haven role in the USA. Generally, depending on the stock market types, time spans, and investment horizons, Bitcoin has different roles in stock portfolios⁹⁰. Finally, only market risk, size, and Bitcoin return in the four-factor model can explain the portfolio excess return, and HML appears to be redundant because it is probably explained by the other three factors in the model. It would be better to have only three factors (market risk premium, SMB, and bitcoin return) if we want to examine the power of bitcoin in explaining the variations of stock portfolios' excess returns in the context of the Fama and French three-factor model. After removing the value factor from the model, we can see that the joint explanatory power (adjusted R²) of other factors is reduced by only 0.14%. That shows that the value factor is practically ineffective in the four-factor model in this research. The coefficients of Lmarket risk premium, LSMB, and Lbitcoin return change respectively to 0.41, 0.10, and 0.11. The output of the POLS and FGLS of the new model is in [Appendix F, Tables 10 and 11](#).

The results of this research are more like the results of [Cheng et al \(2019\)](#) rather than [Cahill et al. \(2020\)](#) except that HML is insignificant in my model. One of the reasons why Cahill et al.

⁹⁰ [Mariana et al. \(2021\), p. 2.](#)

have different results is that they used the Fama-French five-factor model in their study, but Cheng et al. and I used the three-factor model, and this difference in the number of factors can subject to different results⁹¹. Another reason is the different number of firms in the samples among these two studies and my thesis as Cahill et al. mention in their study.

5.3.4. Diagnostic Tests for Panel Data Regression

The Levin-Lin-Chu unit root test shows that all the variables in this study are stationary, so, we can run the panel data regression. The VIF test shows that there is no multicollinearity in the model because the VIF is for each independent variable less than 10. We have no autocorrelation, but heteroskedasticity appears in the model. To tackle this problem, I use FGLS regression in Stata, so that we can rely on the coefficients and the p-values.

To know how Fama and French's three-factor model works before adding Bitcoin returns only for these 29 companies, the following additional analyses are performed from July 2018 to December 2022. The GRS test in [Appendix F, Table 13](#), shows that the mean absolute alpha is very close to zero, and there is no abnormal return in the regression model and the three factors can capture all the variations in the excess returns of the portfolios. The panel regression shows that the coefficients of all three factors are statistically significant and positive, and the adjusted R-squared of the model is 27.39% which is far less than the explanatory powers in step one (monthly intervals from 2003 to 2022) and step two (daily intervals for all firms from 2018 to 2022). In the first and second steps, the number of firms in the samples was large enough to get more reliable results, but we have only 29 companies in this section and these companies may have the same characteristics. In addition, COVID-19, which covers a fifth of the research period, can have an impact on the regression output. These could be the main reasons why the explanatory power decreases sharply. The regression output is in [Appendix F, Table 14](#). The variables are stationary, and no multicollinearity is detected. To solve the autocorrelation and heteroskedasticity problems I used the FGLS model which can be seen in [Appendix F, Table 15](#).

6. Conclusion and Limitations of Research

In this capital, I summarize the most important results of my thesis and the data which I used. In addition, I will mention the limitations, which I faced in this study.

⁹¹ [Cahill et al. \(2020\), p. 15](#).

6.1. Conclusion

In this thesis, I answer four questions:

- 1- Can the risk factors (market, size, and book-to-market ratio) collectively explain and significantly capture the monthly stock/portfolio excess returns within the S&P Composite 1500 from July 2003 to December 2022?
- 2- Can the risk factors (market, size, and book-to-market ratio) collectively explain and significantly capture the daily stock/portfolio excess returns within the S&P Composite 1500 from July 2018 to December 2022?
- 3- Which portfolio can achieve a higher average return in both periods?
- 4- Can the return of Bitcoin and the other three factors significantly explain the variations in stock/portfolio excess returns within the Fama and French three-factor model from July 2018 to December 2022?

To answer these questions, I performed time-series cross-sectional regressions for the panel data, namely Pooled Ordinary Least Squares (POLS). The period of study is divided into long-term with monthly intervals from July 2003 to December 2022, and short-term daily intervals from July 2018 to December 2022. The selected companies are from the S&P Composite 1500 which covers 90% of the U.S. market capitalization, and all the constructed portfolios in this study are value-weighted. For the fourth research question, I selected only blockchain-based firms to prevent probable biases, which resulted in 29 firms.

The size, value, and market risk premium factors have been proven to explain the excess returns of stocks both in the long term and the short term. The coefficients of the factors were highly significant, and the Gibbons-Ross-Shanken (GRS) test indicated that these three risk factors capture all the variations in stock portfolio excess returns both in the long term and short term. The explanatory power of the factors for the monthly data in the period from July 2003 to December 2022 was 79.58% and for the daily data in the period from July 2018 to December 2022 was 71.15%. Furthermore, from the regression outputs, I figured out that the coefficients and R-squared in asset pricing models change in various intervals and periods. This result is consistent with the results of [Fama and French 1993](#) and [Bera et al. \(2020\)](#). Generally, the smaller portfolios have on average higher returns than big portfolios. In the long- and short-run, small-medium and small-high portfolios had respectively the highest average portfolio returns. Small stocks have less liquidity. Therefore, the low level of liquidity makes small stocks more sensitive to market changes and experience larger price fluctuations with the increase of buyers and sellers. Additionally, small firms have a higher cost of capital than larger companies; in other words, they are riskier, and because of their high-risk tolerance, compared to large

companies, they have more returns. Stocks with a higher book-to-market ratio may have higher returns because investors think that the market price of these stocks is lower than their book value, and in the future, their prices will increase and they will buy them, which leads to an increase in demand for stocks. Finally, we observe the increase in stock prices and returns. We concluded from equation 3.4, that stocks with higher BE/ME have higher returns.

In the test for the 29 blockchain-based firms, value appeared to be not a significant factor in explaining the variations of portfolio excess returns after adding the returns of Bitcoin to the model. However, like [Cheng et al. \(2019\)](#) and in contrast to [Cahill et al. \(2020\)](#), the other three factors had statistically significant coefficients. The adjusted R-squared of the model was 19.31% and the coefficient of the Bitcoin return was 0.13 which is a sign of the diversification role of Bitcoin in stock portfolios according to [Huang et al. \(2021\)](#) and [Chan et al. \(2019\)](#). The interpretation of the coefficients of the four-factor model is as follows:

The logarithm of SMB (Small Minus Big): A one-unit increase in the logarithm of SMB is associated with a 0.21 increase in the logarithm of portfolio excess returns, *ceteris paribus*.

The Logarithm of Market Risk Premium: A one-unit increase in the logarithm of the market risk premium is associated with a 0.37 increase in the logarithm of portfolio excess returns, *ceteris paribus*.

The logarithm of Bitcoin Return: A one-unit increase in the logarithm of Bitcoin return is associated with a 0.13 increase in the logarithm of portfolio excess returns, *ceteris paribus*.

The Logarithm of HML (High Minus Low): The coefficient is -0.04, and as it is statistically not significant, it is impossible to meaningfully interpret HML in this model. In this specific analysis, its effect on the logarithm of portfolio excess returns may not be statistically significant.

According to the results, it can be concluded that all four hypotheses of the study have been proven and accepted.

6.2. Limitations of Research

This research has several limitations. First, I did not exclude the effect of important events like the financial crisis and the COVID-19 pandemic which affect the results according to [Horvath and Wang \(2021\)](#), who concluded in their study that R^2 and the coefficients change due to these events.

Second, it should be noted that the cryptocurrency market is relatively new and has different behaviors from the stock market which is a mature market. As this market, especially Bitcoin,

becomes more accepted in the upcoming years different results might be achieved if this research is conducted again in the future.

Third, the number of blockchain firms is relatively low. We know that more accurate results are obtained by increasing the number of observations in research. However, in this research, there was no solution for this issue.

For future studies, I recommend that the researchers choose another reference market both in the U.S. and out of the U.S. market to compare the results of their study with my thesis. The researchers can also use cryptocurrencies other than Bitcoin, like Ethereum, Tether, Binance Coin, etc.

Appendix

A Overall Cryptocurrency Market Capitalization per Week

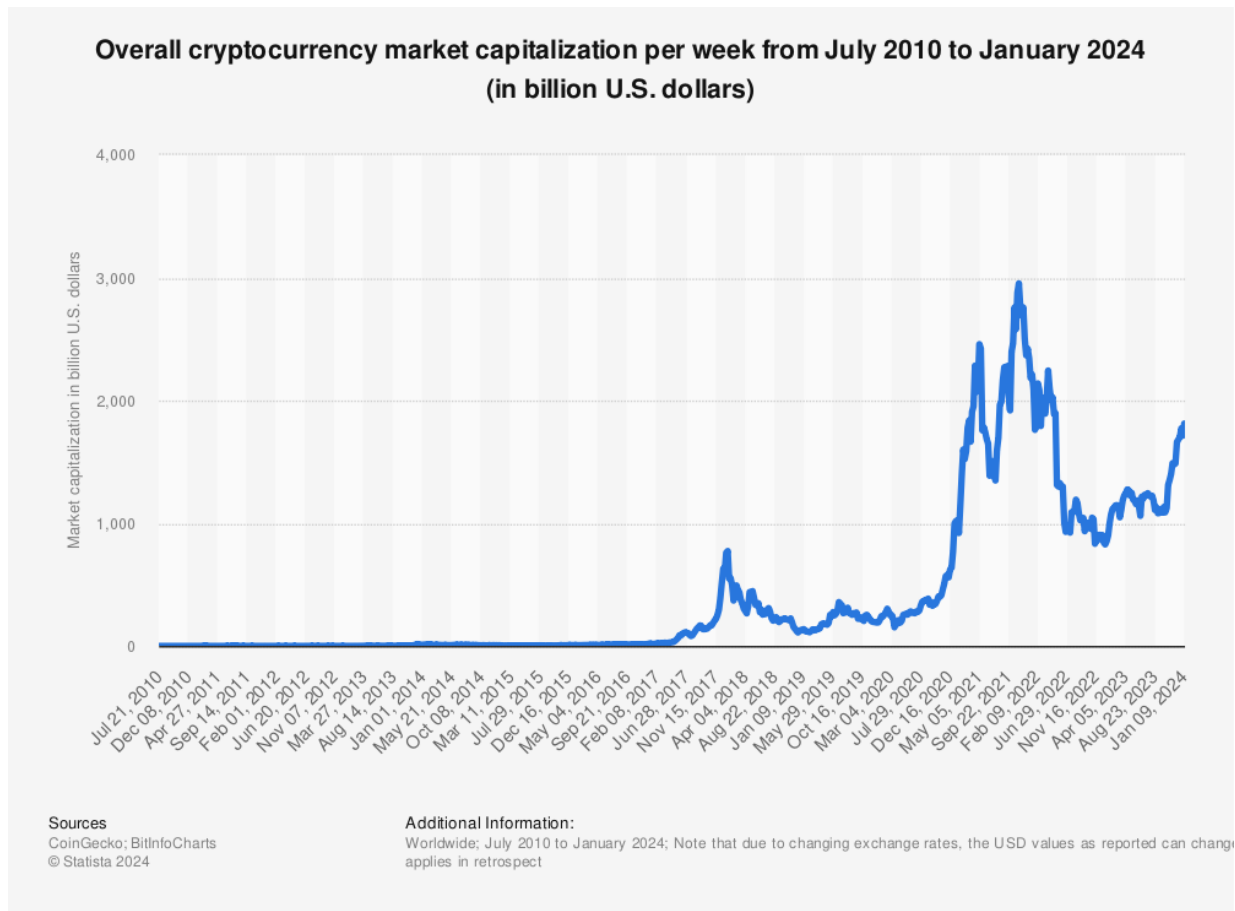


Figure A1 Overall cryptocurrency market capitalization per week from July 2010 to January 2024 (in billion U.S. dollars). See <https://www.statista.com/statistics/730876/cryptocurrency-market-value/> [last accessed on 23.02.2024]

B Market Capitalization per Currency as of 5. December 2023

	Crypto category	Market cap in billion U.S. dollars	Market cap ratio
Bitcoin (BTC)	Store of value	813.96	54.39%
Ethereum (ETH)	Smart contracts	264.80	17.69%
Tether USDt (USDT)	Stablecoin	89.80	6.00%
Binance Coin (BNB)	Exchange token (centralized)	34.83	2.33%
Ripple (XRP)	Payments/digital currency	32.99	2.20%
Solana (SOL)	Smart contracts	25.44	1.70%
USDC (USDC)	Stablecoin	24.31	1.62%
Cardano (ADA)	Smart contracts	14.13	0.94%
Dogecoin (DOGE)	Memecoin	12.54	0.84%
TRON (TRX)	Payments/digital currency	9.09	0.61%
Chainlink (LINK)	Exchange token (decentralized)/DeFi	8.53	0.57%
Toncoin (TON)	Payments/digital currency	8.18	0.55%
Avalanche (AVAX)	Exchange token (decentralized)/DeFi	8.10	0.54%
Polkadot (DOT)	Smart contracts	7.36	0.49%
Polygon (MATIC)	Payments/digital currency	7.33	0.49%
Dai (DAI)	Stablecoin	5.34	0.36%
Litecoin (LTC)	Payments/digital currency	5.32	0.36%
Shiba Inu (SHIB)	Memecoin	5.28	0.35%
Bitcoin Cash (BCH)	Payments/digital currency	4.71	0.31%
Cosmos (ATOM)	Interoperability	3.62	0.24%
Uniswap (UNI)	Exchange token (decentralized)/DeFi	3.55	0.24%
OKB (OKB)	Exchange token (centralized)	3.47	0.23%
UNUS SED LEO (LEO)	Exchange token (centralized)	3.45	0.23%

Stellar (XLM)	Exchange token (decentralized)/DeFi	3.39	0.23%
Monero (XMR)	Payments/digital currency	3.11	0.21%
Kaspa (KAS)	Store of value	3.06	0.20%
TrueUSD (TUSD)	Stablecoin	2.85	0.19%
Ethereum Classic (ETC)	Smart contracts	2.82	0.19%
Cronos (CRO)	Exchange token (centralized)	2.38	0.16%
Internet Computer (ICP)	Distributed computing	2.25	0.15%
NEAR Protocol (NEAR)	Smart contracts	2.21	0.15%
Filecoin (FIL)	Smart contracts	2.19	0.15%
Hedera (HBAR)	Payments/digital currency	2.16	0.14%
THORChain (RUNE)	Exchange token (decentralized)/DeFi	2.13	0.14%
Lido DAO (LDO)	Exchange token (decentralized)/DeFi	2.10	0.14%
Aptos (APT)	Smart contracts	2.07	0.14%
Immutable (IMX)	Exchange token (centralized)	1.85	0.12%
Mantle (MNT)	Interoperability	1.77	0.12%
VeChain (VET)	Smart contracts	1.70	0.11%
BUSD (BUSD)	Stablecoin	1.62	0.11%
Optimism (OP)	Interoperability	1.58	0.11%
Stacks (STX)	Exchange token (decentralized)/DeFi	1.57	0.10%
FTX Token (FTT)	Exchange token (centralized)	1.43	0.10%
Aave (AAVE)	Exchange token (decentralized)/DeFi	1.42	0.09%

The Graph (GRT)	Exchange token (decentralized)/DeFi	1.41	0.09%
Terra Classic (LUNC)	Stablecoin	1.39	0.09%
Injective (INJ)	Exchange token (decentralized)/DeFi	1.36	0.09%
Arbitrum (ARB)	Distributed computing	1.36	0.09%
Maker (MKR)	Exchange token (decentralized)/DeFi	1.35	0.09%
Render (RNDR)	NFT tokens	1.31	0.09%
MultiversX (EGLD)	Smart contracts	1.31	0.09%
Celestia (TIA)	Interoperability	1.27	0.08%
Quant (QNT)	Interoperability	1.22	0.08%
Algorand (ALGO)	Exchange token (centralized)	1.21	0.08%
Flow (FLOW)	Payments/digital currency	1.20	0.08%
Theta Network (THETA)	NFT tokens	1.13	0.08%
Synthetix (SNX)	Exchange token (decentralized)/DeFi	1.07	0.07%
ORDI (ORDI)	NFT tokens	1.04	0.07%
Bitcoin SV (BSV)	Store of value	0.98	0.07%
First Digital USD (FDUSD)	Stablecoin	0.97	0.06%
IOTA (IOTA)	Exchange token (decentralized)/DeFi	0.96	0.06%
The Sandbox (SAND)	NFT tokens	0.93	0.06%
KuCoin Token (KCS)	Exchange token (centralized)	0.92	0.06%
Axie Infinity (AXS)	NFT tokens	0.92	0.06%
Fantom (FTM)	Memecoin	0.89	0.06%

Decentraland (MANA)	Payments/digital currency	0.88	0.06%
Neo (NEO)	Smart contracts	0.84	0.06%
Tezos (XTZ)	Exchange token (decentralized)/DeFi	0.84	0.06%
Kava (KAVA)	Exchange token (decentralized)/DeFi	0.81	0.05%
Gala (GALA)	Exchange token (centralized)	0.80	0.05%
EOS (EOS)	Smart contracts	0.80	0.05%
Mina (MINA)	Smart contracts	0.75	0.05%
Conflux (CFX)	Interoperability	0.75	0.05%
Bitget Token (BGB)	Exchange token (decentralized)/DeFi	0.72	0.05%
USDD (USDD)	Stablecoin	0.72	0.05%
XDC Network (XDC)	Exchange token (decentralized)/DeFi	0.69	0.05%
Klaytn (KLAY)	Exchange token (centralized)	0.67	0.04%
Terra (LUNA)	Exchange token (decentralized)/DeFi	0.66	0.04%
Sui (SUI)	Smart contracts	0.65	0.04%
Frax Share (FXS)	Interoperability	0.64	0.04%
Sei (SEI)	Exchange token (decentralized)/DeFi	0.64	0.04%
Blur (BLUR)	NFT tokens	0.61	0.04%
PancakeSwap (CAKE)	Exchange token (decentralized)/DeFi	0.60	0.04%
Rocket Pool (RPL)	Exchange token (decentralized)/DeFi	0.60	0.04%
eCash (XEC)	Store of value	0.60	0.04%
Chiliz (CHZ)	Payments/digital currency	0.60	0.04%
ApeCoin (APE)	NFT tokens	0.60	0.04%

Pepe (PEPE)	Memecoin	0.57	0.04%
dYdX (ethDYDX) (ETHDYDX)	Exchange token (decentralized)/DeFi	0.57	0.04%
Curve DAO Token (CRV)	Exchange token (decentralized)/DeFi	0.55	0.04%
Oasis Network (ROSE)	Exchange token (decentralized)/DeFi	0.54	0.04%
Arweave (AR)	Distributed computing	0.53	0.04%
Gnosis (GNO)	Exchange token (decentralized)/DeFi	0.51	0.03%
Gas (GAS)	Payments/digital currency	0.51	0.03%
Beam (BEAM)	Exchange token (decentralized)/DeFi	0.51	0.03%
Tether Gold (XAUT)	Stablecoin	0.50	0.03%
Zcash (ZEC)	Smart contracts	0.50	0.03%
Trust Wallet Token (TWT)	Smart contracts	0.50	0.03%
TerraClassicUSD (USTC)	Stablecoin	0.48	0.03%
BitTorrent (New) (BTT)	Smart contracts	0.47	0.03%
PAX Gold (PAXG)	Smart contracts	0.47	0.03%
GMX (GMX)	Exchange token (centralized)	0.46	0.03%
Nexo (NEXO)	Exchange token (decentralized)/DeFi	0.46	0.03%
Huobi Token (HT)	Exchange token (centralized)	0.45	0.03%
Pax Dollar (USDP)	Stablecoin	0.44	0.03%
Fetch.ai (FET)	NFT tokens	0.43	0.03%
GateToken (GT)	Exchange token (centralized)	0.42	0.03%

Compound (COMP)	Exchange token (decentralized)/DeFi	0.42	0.03%
Casper (CSPR)	Smart contracts	0.41	0.03%
WOO Network (WOO)	Exchange token (centralized)	0.41	0.03%
aelf (ELF)	Smart contracts	0.41	0.03%
Flare (FLR)	Payments/digital currency	0.41	0.03%
STEPN (GMT)	Smart contracts	0.41	0.03%
IoTeX (IOTX)	Smart contracts	0.41	0.03%
Illuvium (ILV)	NFT tokens	0.40	0.03%
Zilliqa (ZIL)	Payments/digital currency	0.40	0.03%
Helium (HNT)	Distributed computing	0.39	0.03%
Akash Network (AKT)	Distributed computing	0.39	0.03%
1inch Network (1INCH)	Exchange token (decentralized)/DeFi	0.39	0.03%
SingularityNET (AGIX)	Payments/digital currency	0.38	0.03%

Table B1 Biggest cryptocurrencies in the world - both coins and tokens - based on market capitalization on December 5, 2023 (in billion U.S. dollars) – Source: <https://www.statista.com/statistics/1269013/biggest-crypto-per-category-worldwide/> [last accessed on 06.12.2023]

C List of Blockchain Stocks

Ticker	Name of Compnay	Ticker	Name of Compnay
AMD	Advanced Micro Devices Inc	INTU	Intuit Inc
PLXS	Plexus Corp	FIS	Fidelity National Information Services Inc
CTSH	Cognizant Technology Solutions Corp	NDAQ	Nasdaq Inc
DLX	Deluxe Corp	MAT	Mattel Inc
VMI	Valmont Industries Inc	DGX	Quest Diagnostics Inc
ORCL	Oracle Corp	NVDA	NVIDIA Corp
MA	Mastercard Inc	AKAM	Akamai Technologies Inc
BR	Broadridge Financial Solutions Inc	NEM	Newmont Corporation
NSIT	Insight Enterprises Inc	VYX	NCR Voyix Corp
PLAB	Photronics Inc	PYPL	PayPal Holdings Inc
ACN	Accenture PLC	GME	GameStop Corp
NKE	Nike Inc	MMM	3M Co
V	Visa Inc	PAYC	Paycom Software Inc
BFH	Bread Financial Holdings Inc	TSLA	Tesla Inc
IBM	International Business Machines Corp		

Table C1 U.S. blockchain stocks in S&P Composite 1500 according to 10-K filings

D Statistical Outputs (monthly data from July 2003 to December 2022)

```
. corr SMB smb_ff
(obs=234)
```

	SMB	smb_ff
SMB	1.0000	
smb_ff	0.7947	1.0000

```
. corr HML hml_ff
(obs=234)
```

	HML	hml_ff
HML	1.0000	
hml_ff	0.8034	1.0000

```
. corr Market_RiskPremium mktrf_ff
(obs=234)
```

	Market_RiskPremium	mktrf_ff
Market_RiskPremium	1.0000	
mktrf_ff	0.9960	1.0000

Table D1 Correlations between the risk factors calculated based on the data from S&P Composite 1500 and the risk factors from Kenneth R. French database (monthly data). The factors on Kenneth R. French website are smb_ff, hml_ff, and mktrf_ff.

```
. ttest vwret == 0 if portfolio == "small low"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0071981	.0032486	.0496935	.0007978	.0135984

```
mean = mean(vwret)
H0: mean = 0
t = 2.2158
Degrees of freedom = 233
```

```
Ha: mean < 0
Pr(T < t) = 0.9862
Ha: mean != 0
Pr(|T| > |t|) = 0.0277
Ha: mean > 0
Pr(T > t) = 0.0138
```

```
. ttest vwret == 0 if portfolio == "small medium"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0075135	.0032435	.049616	.0011232	.0139038

```
mean = mean(vwret)
H0: mean = 0
t = 2.3165
Degrees of freedom = 233
```

```
Ha: mean < 0
Pr(T < t) = 0.9893
Ha: mean != 0
Pr(|T| > |t|) = 0.0214
Ha: mean > 0
Pr(T > t) = 0.0107
```

```
. ttest vwret == 0 if portfolio == "small high"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0072711	.0039454	.0603531	-.0005021	.0150443

```
mean = mean(vwret)                                t = 1.8429
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9667                               Pr(|T| > |t|) = 0.0666                               Pr(T > t) = 0.0333
```

```
. ttest vwret == 0 if portfolio == "big low"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0062855	.0026964	.0412467	.0009731	.011598

```
mean = mean(vwret)                                t = 2.3311
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9897                               Pr(|T| > |t|) = 0.0206                               Pr(T > t) = 0.0103
```

```
. ttest vwret == 0 if portfolio == "big medium"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0057332	.0024822	.0379697	.0008428	.0106235

```
mean = mean(vwret)                                t = 2.3098
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9891                               Pr(|T| > |t|) = 0.0218                               Pr(T > t) = 0.0109
```

```
. ttest vwret == 0 if portfolio == "big high"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	234	.0044231	.0028227	.0431794	-.0011382	.0099845

```
mean = mean(vwret)                                t = 1.5670
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9408                               Pr(|T| > |t|) = 0.1185                               Pr(T > t) = 0.0592
```

Table D2 T-statistics of value-weighted average return for each portfolio (monthly data). The vwret stands for value-weighted average return

```
. regress small_low Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	234
Model	.554399069	3	.18479969	F(3, 230)	=	1984.43
Residual	.021418684	230	.000093125	Prob > F	=	0.0000
				R-squared	=	0.9628
				Adj R-squared	=	0.9623
Total	.575817753	233	.002471321	Root MSE	=	.00965

small_low	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.9826519	.0181105	54.26	0.000	.9469682	1.018336
SMB	.975608	.0323005	30.20	0.000	.9119652	1.039251
HML	-.4345282	.0305079	-14.24	0.000	-.4946389	-.3744175
_cons	-.0011723	.0006401	-1.83	0.068	-.0024336	.000089

```
. regress small_medium Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	234
Model	.549832909	3	.183277636	F(3, 230)	=	1894.14
Residual	.02225492	230	.000096761	Prob > F	=	0.0000
				R-squared	=	0.9611
				Adj R-squared	=	0.9606
Total	.572087828	233	.002455313	Root MSE	=	.00984

small_medium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.9456628	.0184607	51.23	0.000	.9092892	.9820364
SMB	.8523585	.0329251	25.89	0.000	.7874852	.9172317
HML	-.0123566	.0310977	-0.40	0.691	-.0736295	.0489163
_cons	-.0001299	.0006525	-0.20	0.842	-.0014155	.0011558

```
. regress small_high Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	234
Model	.822505728	3	.274168576	F(3, 230)	=	2071.59
Residual	.030439731	230	.000132347	Prob > F	=	0.0000
				R-squared	=	0.9643
				Adj R-squared	=	0.9638
Total	.852945459	233	.00366071	Root MSE	=	.0115

small_high	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.062493	.0215901	49.21	0.000	1.019953	1.105032
SMB	.9016831	.0385065	23.42	0.000	.8258126	.9775537
HML	.4816309	.0363694	13.24	0.000	.4099711	.5532907
_cons	-.0005388	.0007631	-0.71	0.481	-.0020425	.0009648

. regress big_low Market_RiskPremium SMB HML

Source	SS	df	MS	Number of obs	=	234
Model	.374526617	3	.124842206	F(3, 230)	=	1201.44
Residual	.023899353	230	.00010391	Prob > F	=	0.0000
				R-squared	=	0.9400
				Adj R-squared	=	0.9392
Total	.39842597	233	.001709983	Root MSE	=	.01019

big_low	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.094493	.0191305	57.21	0.000	1.0568	1.132187
SMB	-.0854822	.0341198	-2.51	0.013	-.1527095	-.0182549
HML	-.4175723	.0322262	-12.96	0.000	-.4810686	-.354076
_cons	-.00048	.0006762	-0.71	0.479	-.0018123	.0008523

. regress big_medium Market_RiskPremium SMB HML

Source	SS	df	MS	Number of obs	=	234
Model	.315090695	3	.105030232	F(3, 230)	=	1053.45
Residual	.022931244	230	.000099701	Prob > F	=	0.0000
				R-squared	=	0.9322
				Adj R-squared	=	0.9313
Total	.338021939	233	.001450738	Root MSE	=	.00999

big_medium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.987951	.0187391	52.72	0.000	.9510288	1.024873
SMB	-.1179085	.0334216	-3.53	0.001	-.1837601	-.0520568
HML	-.0043545	.0315667	-0.14	0.890	-.0665514	.0578425
_cons	-.00002	.0006624	-0.03	0.976	-.0013251	.0012851

. regress big_high Market_RiskPremium SMB HML

Source	SS	df	MS	Number of obs	=	234
Model	.407874271	3	.13595809	F(3, 230)	=	1193.26
Residual	.026205748	230	.000113938	Prob > F	=	0.0000
				R-squared	=	0.9396
				Adj R-squared	=	0.9388
Total	.43408002	233	.001863004	Root MSE	=	.01067

big_high	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.9475342	.0200324	47.30	0.000	.9080638	.9870047
SMB	-.0055158	.0357282	-0.15	0.877	-.0759123	.0648807
HML	.5637371	.0337454	16.71	0.000	.4972476	.6302267
_cons	-.0007385	.0007081	-1.04	0.298	-.0021336	.0006566

Table D3 Time-series regression outputs for each portfolio (monthly data). Variables small_low, small_medium, small_high, big_low, big_medium, and big_high are portfolios excess returns

```
. grsftest big_high-small_medium, factor(Market_RiskPremium) d
```

GRS Test Results

```
Number of Obs. :      234
Number of Asset :       6
Number of Factor:      1
Results
      GRS_F-test = 0.738132
      GRS_pvalue = 0.619416
```

```
Amean[1,4]: Factor Model Results (On Average)
```

	Mean alpha	Mean SE	Mean R2	Mean R2_adj
6 Assets	-0.000269	0.001259	0.839297	0.838605

```
. grsftest big_high-small_medium, factor(Market_RiskPremium SMB) d
```

GRS Test Results

```
Number of Obs. :      234
Number of Asset :       6
Number of Factor:      2
Results
      GRS_F-test = 0.789378
      GRS_pvalue = 0.579062
```

```
Amean[1,4]: Factor Model Results (On Average)
```

	Mean alpha	Mean SE	Mean R2	Mean R2_adj
6 Assets	-0.000577	0.000852	0.920488	0.919799

```
. grsftest big_high-small_medium, factor(Market_RiskPremium HML) d
```

GRS Test Results

```
Number of Obs. :      234
Number of Asset :       6
Number of Factor:      2
Results
      GRS_F-test = 0.583634
      GRS_pvalue = 0.743228
```

```
Amean[1,4]: Factor Model Results (On Average)
```

	Mean alpha	Mean SE	Mean R2	Mean R2_adj
6 Assets	0.000123	0.001027	0.891455	0.890516


```
. grsftest big_high-small_medium, factor(SMB HML Market_RiskPremium) d
```

GRS Test Results

```
Number of Obs. :      234
Number of Asset :       6
Number of Factor:      3
Results
      GRS_F-test = 0.697097
      GRS_pvalue = 0.652217
```

```
Amean[1,4]: Factor Model Results (On Average)
```

	Mean alpha	Mean SE	Mean R2	Mean R2_adj
6 Assets	-0.000513	0.000684	0.950003	0.949351

Table D4 GRS-test outputs (monthly data)

Outputs of data from Kenneth R. French database:

```
. ttest vwre == 0 if portfolio == "SMALL LoBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	.8387303	.3337457	5.105327	.1811854	1.496275

```
mean = mean(vwre)                                t =    2.5131
H0: mean = 0                                     Degrees of freedom =    233
```

Ha: mean < 0	Ha: mean != 0	Ha: mean > 0
Pr(T < t) = 0.9937	Pr(T > t) = 0.0126	Pr(T > t) = 0.0063

```
. ttest vwre == 0 if portfolio == "SMALL MeBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	1.008071	.3145673	4.811954	.3883113	1.627831

```
mean = mean(vwre)                                t =    3.2046
H0: mean = 0                                     Degrees of freedom =    233
```

Ha: mean < 0	Ha: mean != 0	Ha: mean > 0
Pr(T < t) = 0.9992	Pr(T > t) = 0.0015	Pr(T > t) = 0.0008

```
. ttest vwre == 0 if portfolio == "SMALL HiBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	1.057842	.3483457	5.328665	.371532	1.744152

```
mean = mean(vwre)                                t = 3.0368
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                Ha: mean != 0                Ha: mean > 0
Pr(T < t) = 0.9987          Pr(|T| > |t|) = 0.0027          Pr(T > t) = 0.0013
```

```
. ttest vwre == 0 if portfolio == "BIG LoBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	.9240671	.2456325	3.757454	.4401226	1.408012

```
mean = mean(vwre)                                t = 3.7620
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                Ha: mean != 0                Ha: mean > 0
Pr(T < t) = 0.9999          Pr(|T| > |t|) = 0.0002          Pr(T > t) = 0.0001
```

```
. ttest vwre == 0 if portfolio == "BIG MeBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	.8590436	.2426766	3.712239	.3809227	1.337165

```
mean = mean(vwre)                                t = 3.5399
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                Ha: mean != 0                Ha: mean > 0
Pr(T < t) = 0.9998          Pr(|T| > |t|) = 0.0005          Pr(T > t) = 0.0002
```

```
. ttest vwre == 0 if portfolio == "BIG HiBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	234	1.026073	.2985585	4.567067	.4378539	1.614292

```
mean = mean(vwre)                                t = 3.4368
H0: mean = 0                                     Degrees of freedom = 233
```

```
Ha: mean < 0                Ha: mean != 0                Ha: mean > 0
Pr(T < t) = 0.9997          Pr(|T| > |t|) = 0.0007          Pr(T > t) = 0.0003
```

Table D5 T-statistics of value-weighted average return for each portfolio (monthly data from Kenneth R. French database). The vwre stands for value-weighted average return

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL LoBM"
```

Source	SS	df	MS	Number of obs	=	234
Model	8295.47728	3	2765.15909	F(3, 230)	=	1118.38
Residual	568.669642	230	2.47247671	Prob > F	=	0.0000
				R-squared	=	0.9358
				Adj R-squared	=	0.9350
Total	8864.14692	233	38.043549	Root MSE	=	1.5724

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.152217	.0278617	41.35	0.000	1.097321	1.207114
SMB	1.241022	.0484878	25.59	0.000	1.145485	1.336559
HML	-.2637506	.0422109	-6.25	0.000	-.3469202	-.1805811
_cons	-.1514978	.1048471	-1.44	0.150	-.3580815	.0550858

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL MeBM"
```

Source	SS	df	MS	Number of obs	=	234
Model	7016.94046	3	2338.98015	F(3, 230)	=	1593.14
Residual	337.676374	230	1.46815815	Prob > F	=	0.0000
				R-squared	=	0.9541
				Adj R-squared	=	0.9535
Total	7354.61683	233	31.5648791	Root MSE	=	1.2117

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.085643	.0214698	50.57	0.000	1.043341	1.127946
SMB	.9458143	.0373639	25.31	0.000	.872195	1.019434
HML	.3505759	.0325271	10.78	0.000	.2864868	.4146651
_cons	-.0276769	.0807936	-0.34	0.732	-.1868671	.1315133

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL HiBM"
```

Source	SS	df	MS	Number of obs	=	234
Model	8744.31255	3	2914.77085	F(3, 230)	=	1458.49
Residual	459.65128	230	1.99848383	Prob > F	=	0.0000
				R-squared	=	0.9501
				Adj R-squared	=	0.9494
Total	9203.96383	233	39.5019907	Root MSE	=	1.4137

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.107748	.0250491	44.22	0.000	1.058393	1.157103
SMB	1.009843	.043593	23.17	0.000	.9239501	1.095735
HML	.8722088	.0379498	22.98	0.000	.7974352	.9469824
_cons	-.0701227	.0942629	-0.74	0.458	-.2558519	.1156065

. regress vwreRF MktRF SMB HML if portfolio == "BIG LoBM"

Source	SS	df	MS	Number of obs	=	234
Model	3991.61781	3	1330.53927	F(3, 230)	=	1477.20
Residual	207.164516	230	.900715288	Prob > F	=	0.0000
				R-squared	=	0.9507
				Adj R-squared	=	0.9500
Total	4198.78233	233	18.020525	Root MSE	=	.94906

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.075019	.0168165	63.93	0.000	1.041885	1.108153
SMB	-.0838479	.0292658	-2.87	0.005	-.1415112	-.0261846
HML	-.2946187	.0254773	-11.56	0.000	-.3448173	-.24442
_cons	.0342002	.0632826	0.54	0.589	-.0904875	.1588879

. regress vwreRF MktRF SMB HML if portfolio == "BIG MeBM"

Source	SS	df	MS	Number of obs	=	234
Model	4179.13328	3	1393.04443	F(3, 230)	=	695.91
Residual	460.406128	230	2.00176577	Prob > F	=	0.0000
				R-squared	=	0.9008
				Adj R-squared	=	0.8995
Total	4639.53941	233	19.9121863	Root MSE	=	1.4148

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.044484	.0250697	41.66	0.000	.9950886	1.09388
SMB	.0012608	.0436288	0.03	0.977	-.0847024	.0872239
HML	.3004265	.0379809	7.91	0.000	.2255915	.3752615
_cons	-.1461224	.0943403	-1.55	0.123	-.3320041	.0397592

. regress vwreRF MktRF SMB HML if portfolio == "BIG HiBM"

Source	SS	df	MS	Number of obs	=	234
Model	7150.92871	3	2383.6429	F(3, 230)	=	640.58
Residual	855.840234	230	3.7210445	Prob > F	=	0.0000
				R-squared	=	0.8931
				Adj R-squared	=	0.8917
Total	8006.76895	233	34.3638152	Root MSE	=	1.929

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.203817	.0341802	35.22	0.000	1.13647	1.271163
SMB	.0910105	.0594838	1.53	0.127	-.0261923	.2082133
HML	.9430967	.0517835	18.21	0.000	.8410661	1.045127
_cons	-.1733229	.1286243	-1.35	0.179	-.4267554	.0801096

Table D6 Time-series regression outputs for each portfolio (Kenneth R. French monthly data). The vwreRF and MktRF are respectively portfolio value-weighted average return minus risk-free rate and market return minus risk-free rate

Breusch and Pagan Lagrangian multiplier test for random effects

$\text{vwret_risk_premium}[\text{eportfolio}, t] = Xb + u[\text{eportfolio}] + e[\text{eportfolio}, t]$

Estimated results:

	Var	SD = sqrt(Var)
vwret_r~m	.0033803	.0581401
e	.0006912	.0262915
u	0	0

Test: $\text{Var}(u) = 0$

chibar2(01) = 0.00
Prob > chibar2 = 1.0000

Table D7 Breusch and Pagan Lagrangian multiplier test for random effects (monthly data)

. regress vwret_risk_premium Market_RiskPremium SMB HML

Source	SS	df	MS	Number of obs	=	1,404
Model	3.77639344	3	1.25879781	F(3, 1400)	=	1824.11
Residual	.96612308	1,400	.000690088	Prob > F	=	0.0000
				R-squared	=	0.7963
				Adj R-squared	=	0.7958
Total	4.74251652	1,403	.003380268	Root MSE	=	.02627

vwret_risk_premium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.161442	.0201268	57.71	0.000	1.12196	1.200924
SMB	.4975092	.0358967	13.86	0.000	.4270922	.5679262
HML	.1327445	.0339044	3.92	0.000	.0662356	.1992535
_cons	-.0017516	.0007114	-2.46	0.014	-.0031471	-.0003561

Table D8 Panel regression output (monthly data)

Cross-sectional time-series FGLS regression

Coefficients: generalized least squares

Panels: homoskedastic

Correlation: no autocorrelation

Estimated covariances	=	1	Number of obs	=	1,404
Estimated autocorrelations	=	0	Number of groups	=	6
Estimated coefficients	=	4	Time periods	=	234
			Wald chi2(3)	=	5487.97
Log likelihood	=	3119.455	Prob > chi2	=	0.0000

vwret_risk_premium	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Market_RiskPremium	1.161442	.0200981	57.79	0.000	1.122051	1.200834
SMB	.4975092	.0358455	13.88	0.000	.4272534	.5677651
HML	.1327445	.0338561	3.92	0.000	.0663878	.1991012
_cons	-.0017516	.0007104	-2.47	0.014	-.0031439	-.0003593

Table D9 FGLS model (monthly data)

```
. dfuller SMB, noconstant lags(1)
```

[illegible]

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.523	-2.582	-1.950	-1.619

[illegible]

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.046	-2.582	-1.950	-1.619

```
Variable: Market_RiskPremium      Number of obs = 232
                                   Number of lags = 1
```

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.673	-2.582	-1.950	-1.619

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.608	-2.582	-1.950	-1.619

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.813	-2.582	-1.950	-1.619

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-11.131	-2.582	-1.950	-1.619

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.964	-2.582	-1.950	-1.619

Table D10 Augmented Dickey-Fuller test for stationary (unit root) test (monthly data)


```
. vif
```

Variable	VIF	1/VIF
SMB	1.56	0.642052
HML	1.38	0.725771
Market_Ris~m	1.21	0.828432
Mean VIF	1.38	

Table D11 Variance Inflation Factor test for Multicollinearity (monthly data)

Small-low portfolio:

```
. hettest, rhs fstat
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 230) = 1.61
Prob > F = 0.1875

Table D12 Breusch-Pagan test for Heteroskedasticity (small-low portfolio-monthly data)

```
. estat dwatson
```

Durbin-Watson d-statistic(4, 234) = 2.061248

Table D13 Durbin-Watson test for Autocorrelation (small-low portfolio-monthly data)

```
. jb resid
```

Jarque-Bera normality test: 3.01 Chi(2) .222
Jarque-Bera test for Ho: normality:

Table D14 Jarque-Bera test for Normality (small-low portfolio-monthly data)

Small-medium portfolio:

```
. hettest, rhs fstat

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 230) = 1.67
Prob > F = 0.1743
```

Table D15 Breusch-Pagan test for Heteroskedasticity (small-medium portfolio-monthly data)

```
. estat dwatson

Durbin-Watson d-statistic( 4, 234) = 2.01526
```

Table D16 Durbin-Watson test for Autocorrelation (small-medium portfolio-monthly data)

```
. jb resid
Jarque-Bera normality test: .8397 Chi(2) .6571
Jarque-Bera test for Ho: normality:
```

Table D17 Jarque-Bera test for Normality (small-medium portfolio-monthly data)**Small-high portfolio:**

```
. hettest, rhs fstat

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 230) = 1.71
Prob > F = 0.1659
```

Table D18 Breusch-Pagan test for Heteroskedasticity (small-high portfolio-monthly data)

```
. estat dwatson

Durbin-Watson d-statistic( 4, 234) = 2.291977
```

Table D19 Durbin-Watson test for Autocorrelation (small-high portfolio-monthly data)

```
. jb resid
Jarque-Bera normality test: 38.57 Chi(2) 4.2e-09
Jarque-Bera test for Ho: normality:
```

Table D20 Jarque-Bera test for Normality (small-high portfolio-monthly data)

Big-low portfolio:

```
. hettest, rhs fstat

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 230) = 1.21
Prob > F = 0.3054
```

Table D21 Breusch-Pagan test for Heteroskedasticity (big-low portfolio-monthly data)

```
. estat dwatson

Durbin-Watson d-statistic( 4, 234) = 2.119557
```

Table D22 Durbin-Watson test for Autocorrelation (big-low portfolio-monthly data)

```
. jb resid
Jarque-Bera normality test: 3.674 Chi(2) .1593
Jarque-Bera test for Ho: normality:
```

Table D23 Jarque-Bera test for Normality (big-low portfolio-monthly data)

Big-medium portfolio:

```
. hettest, rhs fstat

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 230) = 1.58
Prob > F = 0.1946
```

Table D24 Breusch-Pagan test for Heteroskedasticity (big-medium portfolio-monthly data)

```
. estat dwatson
```

```
Durbin-Watson d-statistic( 4, 234) = 2.170839
```

Table D25 Durbin-Watson test for Autocorrelation (big-medium portfolio-monthly data)

```
. jb resid
```

```
Jarque-Bera normality test: 3.66 Chi(2) .1604
```

```
Jarque-Bera test for Ho: normality:
```

Table D26 Jarque-Bera test for Normality (big-medium portfolio-monthly data)

Big-high portfolio:

```
. hettest, rhs fstat
```

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
```

```
Assumption: i.i.d. error terms
```

```
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 230) = 3.69
```

```
Prob > F = 0.0126
```

Table D27 Breusch-Pagan test for Heteroskedasticity (big-high portfolio-monthly data)

```
. estat dwatson
```

```
Durbin-Watson d-statistic( 4, 234) = 2.13371
```

Table D28 Durbin-Watson test for Autocorrelation (big-high portfolio-monthly data)

```
. jb resid
```

```
Jarque-Bera normality test: 7.856 Chi(2) .0197
```

```
Jarque-Bera test for Ho: normality:
```

Table D29 Jarque-Bera test for Normality (big-high portfolio-monthly data)

Diagnostic tests for panel regression (POLS):**Levin-Lin-Chu unit-root test for Market_RiskPremium**

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	234
AR parameter: Common		
Panel means: Included	Asymptotics: N/T -> 0	
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 19.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-26.9159	
Adjusted t*	-23.7890	0.0000

Levin-Lin-Chu unit-root test for SMB

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	234
AR parameter: Common		
Panel means: Included	Asymptotics: N/T -> 0	
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 19.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-26.0890	
Adjusted t*	-22.4335	0.0000

Levin-Lin-Chu unit-root test for HML

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	234
AR parameter: Common		
Panel means: Included	Asymptotics: N/T -> 0	
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 19.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-24.7863	
Adjusted t*	-22.5514	0.0000

Levin-Lin-Chu unit-root test for `vwret_risk_premium`

H_0 : Panels contain unit roots Number of panels = 6
 H_a : Panels are stationary Number of periods = 234

AR parameter: Common Asymptotics: N/T -> 0
 Panel means: Included
 Time trend: Not included

ADF regressions: 1 lag
 LR variance: Bartlett kernel, 19.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-27.4685	
Adjusted t*	-24.9714	0.0000

Table D30 Levin-Lin-Chu test for stationary (unit root) test (monthly data)

```
. vif
```

Variable	VIF	1/VIF
SMB	1.56	0.642052
HML	1.38	0.725771
Market_Risk	1.21	0.828432
Mean VIF	1.38	

Table D31 Variance Inflation Factor test for Multicollinearity (monthly data)

```
. hettest, rhs fstat
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
 Assumption: i.i.d. error terms
 Variables: All independent variables

H_0 : Constant variance

$F(3, 1400) = 7.12$
 Prob > F = 0.0001

Table D32 Breusch-Pagan test for Heteroskedasticity (monthly data)

```
. predict resid, residuals
```

```
. jb resid
```

Jarque-Bera normality test: 3.2e+04 Chi(2) 0
 Jarque-Bera test for H_0 : normality:

Table D33 Jarque-Bera test for Normality (monthly data)

```
Wooldridge test for autocorrelation in panel data
H0: no first order autocorrelation
      F( 1,      5) =      0.914
      Prob > F =      0.3830
```

Table D34 Wooldridge test for panel data Autocorrelation (monthly data)

E Statistical Outputs (daily data from July 2018 to December 2022)

	SMB	smb_ff		HML	hml_ff
SMB	1.0000		HML	1.0000	
smb_ff	0.7869	1.0000	hml_ff	0.8366	1.0000

	Market_Ris~m	mktrf_ff
Market_Ris~m	1.0000	
mktrf_ff	0.9962	1.0000

Table E1 Correlations between the risk factors calculated based on the data from S&P Composite 1500 and the risk factors from Kenneth R. French database (daily data). The factors on Kenneth R. French website are smb_ff, hml_ff, and mktrf_ff.

```
. ttest vwret == 0 if portfolio == "small low"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0008389	.0004082	.0137467	.000038	.0016399

```
mean = mean(vwret)                                t =    2.0550
H0: mean = 0                                       Degrees of freedom =    1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9799                             Pr(|T| > |t|) = 0.0401                             Pr(T > t) = 0.0201
```

```
. ttest vwret == 0 if portfolio == "small medium"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0009346	.0004046	.0136235	.0001408	.0017284

```
mean = mean(vwret)                                t =    2.3102
H0: mean = 0                                       Degrees of freedom =    1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.9895                             Pr(|T| > |t|) = 0.0211                             Pr(T > t) = 0.0105
```



```
. ttest vwret == 0 if portfolio == "small high"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0011813	.0004802	.0161701	.0002392	.0021235

```
mean = mean(vwret)                                t = 2.4602
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                Ha: mean != 0                                Ha: mean > 0
Pr(T < t) = 0.9930                        Pr(|T| > |t|) = 0.0140                        Pr(T > t) = 0.0070
```

```
. ttest vwret == 0 if portfolio == "big low"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0007014	.0003809	.0128264	-.0000459	.0014487

```
mean = mean(vwret)                                t = 1.8415
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                Ha: mean != 0                                Ha: mean > 0
Pr(T < t) = 0.9671                        Pr(|T| > |t|) = 0.0658                        Pr(T > t) = 0.0329
```

```
. ttest vwret == 0 if portfolio == "big medium"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0005204	.0003005	.0101201	-.0000692	.00111

```
mean = mean(vwret)                                t = 1.7316
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                Ha: mean != 0                                Ha: mean > 0
Pr(T < t) = 0.9582                        Pr(|T| > |t|) = 0.0836                        Pr(T > t) = 0.0418
```

```
. ttest vwret == 0 if portfolio == "big high"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwret	1,134	.0004938	.0003274	.011026	-.0001486	.0011363

mean = mean(vwret) t = 1.5082
H0: mean = 0 Degrees of freedom = 1133

Ha: mean < 0 Ha: mean != 0 Ha: mean > 0
Pr(T < t) = 0.9341 Pr(|T| > |t|) = 0.1318 Pr(T > t) = 0.0659

Table E2 T-statistics of value-weighted average return for each portfolio (daily data). The vwret stands for value-weighted average return

```
. regress small_low Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.204316135	3	.068105378	F(3, 1130)	=	7887.50
Residual	.009757099	1,130	8.6346e-06	Prob > F	=	0.0000
Total	.214073234	1,133	.000188944	R-squared	=	0.9544
				Adj R-squared	=	0.9543
				Root MSE	=	.00294

small_low	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.03433	.0091507	113.03	0.000	1.016376	1.052284
SMB	.9560339	.0161634	59.15	0.000	.9243202	.9877476
HML	-.3422835	.013436	-25.48	0.000	-.3686458	-.3159211
_cons	.0002201	.0000873	2.52	0.012	.0000487	.0003915

```
. regress small_medium Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.199177661	3	.066392554	F(3, 1130)	=	6524.06
Residual	.011499519	1,130	.000010177	Prob > F	=	0.0000
Total	.21067718	1,133	.000185946	R-squared	=	0.9454
				Adj R-squared	=	0.9453
				Root MSE	=	.00319

small_medium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.019945	.0099343	102.67	0.000	1.000453	1.039437
SMB	.910175	.0175474	51.87	0.000	.8757458	.9446042
HML	-.0146194	.0145865	-1.00	0.316	-.043239	.0140002
_cons	.0003185	.0000948	3.36	0.001	.0001324	.0005045

```
. regress small_high Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.275776615	3	.091925538	F(3, 1130)	=	5025.15
Residual	.020671205	1,130	.000018293	Prob > F	=	0.0000
				R-squared	=	0.9303
				Adj R-squared	=	0.9301
Total	.29644782	1,133	.000261649	Root MSE	=	.00428

small_high	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.125547	.0133192	84.51	0.000	1.099414	1.15168
SMB	.9429038	.0235264	40.08	0.000	.8967434	.9890642
HML	.534494	.0195566	27.33	0.000	.4961226	.5728653
_cons	.000512	.0001271	4.03	0.000	.0002626	.0007615

```
. regress big_low Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.175958988	3	.058652996	F(3, 1130)	=	6272.94
Residual	.010565689	1,130	9.3502e-06	Prob > F	=	0.0000
				R-squared	=	0.9434
				Adj R-squared	=	0.9432
Total	.186524677	1,133	.000164629	Root MSE	=	.00306

big_low	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.13686	.0095224	119.39	0.000	1.118177	1.155544
SMB	-.0547823	.0168199	-3.26	0.001	-.0877839	-.0217806
HML	-.4264432	.0139817	-30.50	0.000	-.4538762	-.3990103
_cons	.0002749	.0000909	3.02	0.003	.0000966	.0004533

```
. regress big_medium Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.109337449	3	.036445816	F(3, 1130)	=	6126.98
Residual	.006721705	1,130	5.9484e-06	Prob > F	=	0.0000
				R-squared	=	0.9421
				Adj R-squared	=	0.9419
Total	.116059155	1,133	.000102435	Root MSE	=	.00244

big_medium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.9674493	.0075951	127.38	0.000	.9525471	.9823514
SMB	-.0893542	.0134157	-6.66	0.000	-.1156766	-.0630317
HML	-.0414654	.0111519	-3.72	0.000	-.0633462	-.0195846
_cons	.0001552	.0000725	2.14	0.032	.000013	.0002975

```
. regress big_high Market_RiskPremium SMB HML
```

Source	SS	df	MS	Number of obs	=	1,134
Model	.128311915	3	.042770638	F(3, 1130)	=	5185.93
Residual	.009319603	1,130	8.2474e-06	Prob > F	=	0.0000
				R-squared	=	0.9323
				Adj R-squared	=	0.9321
Total	.137631519	1,133	.000121475	Root MSE	=	.00287

big_high	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	.9887584	.0089432	110.56	0.000	.9712112	1.006306
SMB	.0295087	.0157969	1.87	0.062	-.0014858	.0605033
HML	.5483526	.0131313	41.76	0.000	.522588	.5741172
_cons	.0000711	.0000854	0.83	0.405	-.0000964	.0002386

Table E3 Time-series regression outputs for each portfolio (daily data). Variables small_low, small_medium, small_high, big_low, big_medium, and big_high are portfolios excess returns

```
. grstest2 big_high-small_medium, flist(Market_RiskPremium) alphas
```

```
R[2,7]
```

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	.0003377	28.153485	.00008791	.78709314	.00017687	.0003377	0
J1	.0003377	4.663283	.00010761	.78709314	.00017687	.0003377	.15766005

```
. grstest2 big_high-small_medium, flist(Market_RiskPremium SMB) alphas
```

```
R[2,7]
```

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	.00025581	33.738695	7.556e-06	.90377634	.00011663	.00025581	0
J1	.00025581	5.5834467	.00001016	.90377634	.00011663	.00025581	.17264909

```
. grstest2 big_high-small_medium, flist(Market_RiskPremium HML) alphas
```

```
R[2,7]
```

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	.00032394	28.147922	.00008812	.87889101	.00013318	.00032394	0
J1	.00032394	4.6582246	.00010902	.87889101	.00013318	.00032394	.15764798

```
. grstest2 big_high-small_medium, flist(Market_RiskPremium SMB HML) alphas
```

```
R[2,7]
```

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	.00025864	34.278245	5.944e-06	.94114968	.00009301	.00025864	0
J1	.00025864	5.6676992	8.174e-06	.94114968	.00009301	.00025864	.17403303

Table E4 GRS-test outputs (daily data)

Outputs of data from Kenneth R. French database:

```
. ttest vwre == 0 if portfolio == "SMALL LoBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.0367108	.0572252	1.927055	-.0755686	.1489901

```
mean = mean(vwre)                                t = 0.6415
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.7393                               Pr(|T| > |t|) = 0.5213                               Pr(T > t) = 0.2607
```

```
. ttest vwre == 0 if portfolio == "SMALL MeBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.032231	.0520771	1.753692	-.0699473	.1344094

```
mean = mean(vwre)                                t = 0.6189
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.7319                               Pr(|T| > |t|) = 0.5361                               Pr(T > t) = 0.2681
```

```
. ttest vwre == 0 if portfolio == "SMALL HiBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.0396032	.0556083	1.872603	-.0695036	.1487099

```
mean = mean(vwre)                                t = 0.7122
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.7618                               Pr(|T| > |t|) = 0.4765                               Pr(T > t) = 0.2382
```

```
. ttest vwre == 0 if portfolio == "BIG LoBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.05194	.0447136	1.505725	-.0357906	.1396707

```
mean = mean(vwre)                                t = 1.1616
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0                                     Ha: mean != 0                                     Ha: mean > 0
Pr(T < t) = 0.8772                               Pr(|T| > |t|) = 0.2456                               Pr(T > t) = 0.1228
```

```
. ttest vwre == 0 if portfolio == "BIG MeBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.0341623	.0426632	1.436681	-.0495456	.1178701

```
mean = mean(vwre)                                t = 0.8007
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0          Ha: mean != 0          Ha: mean > 0
Pr(T < t) = 0.7883    Pr(|T| > |t|) = 0.4234    Pr(T > t) = 0.2117
```

```
. ttest vwre == 0 if portfolio == "BIG HiBM"
```

One-sample t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
vwre	1,134	.055485	.0537276	1.809272	-.0499318	.1609018

```
mean = mean(vwre)                                t = 1.0327
H0: mean = 0                                     Degrees of freedom = 1133
```

```
Ha: mean < 0          Ha: mean != 0          Ha: mean > 0
Pr(T < t) = 0.8490    Pr(|T| > |t|) = 0.3020    Pr(T > t) = 0.1510
```

Table E5 T-statistics of value-weighted average return for each portfolio (daily data from Kenneth R. French database). The vwre stands for value-weighted average return

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL LoBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	2702.82543	3	900.941809	F(3, 1130)	=	6226.02
Residual	163.517612	1,130	.144705852	Prob > F	=	0.0000
				R-squared	=	0.9430
				Adj R-squared	=	0.9428
Total	2866.34304	1,133	2.52987029	Root MSE	=	.3804

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.103126	.0112694	97.89	0.000	1.081015	1.125237
SMB	1.091636	.0190948	57.17	0.000	1.054171	1.129101
HML	-.1692996	.0119163	-14.21	0.000	-.1926802	-.1459189
_cons	.0051672	.0113147	0.46	0.648	-.017033	.0273674

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL MeBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	1902.67151	3	634.223836	F(3, 1130)	=	6621.93
Residual	108.227167	1,130	.095776254	Prob > F	=	0.0000
				R-squared	=	0.9462
				Adj R-squared	=	0.9460
Total	2010.89868	1,133	1.77484437	Root MSE	=	.30948

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.002779	.0091683	109.38	0.000	.9847905	1.020768
SMB	.8485677	.0155346	54.62	0.000	.8180878	.8790476
HML	.3200705	.0096946	33.02	0.000	.3010491	.3390919
_cons	.0097486	.0092051	1.06	0.290	-.0083125	.0278096

```
. regress vwreRF MktRF SMB HML if portfolio == "SMALL HiBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	2129.03604	3	709.67868	F(3, 1130)	=	5779.33
Residual	138.759489	1,130	.122796008	Prob > F	=	0.0000
				R-squared	=	0.9388
				Adj R-squared	=	0.9387
Total	2267.79553	1,133	2.00158476	Root MSE	=	.35042

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	.9922723	.0103813	95.58	0.000	.9719036	1.012641
SMB	.8743132	.0175899	49.71	0.000	.8398007	.9088257
HML	.7067371	.0109772	64.38	0.000	.6851991	.7282751
_cons	.0126692	.010423	1.22	0.224	-.0077814	.0331198

```
. regress vwreRF MktRF SMB HML if portfolio == "BIG LoBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	1478.37369	3	492.791229	F(3, 1130)	=	20264.59
Residual	27.4791734	1,130	.024317853	Prob > F	=	0.0000
				R-squared	=	0.9818
				Adj R-squared	=	0.9817
Total	1505.85286	1,133	1.32908461	Root MSE	=	.15594

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.041952	.0046198	225.54	0.000	1.032887	1.051016
SMB	-.1433331	.0078277	-18.31	0.000	-.1586916	-.1279747
HML	-.2249739	.004885	-46.05	0.000	-.2345585	-.2153892
_cons	-.0063668	.0046383	-1.37	0.170	-.0154675	.002734


```
. regress vwreRF MktRF SMB HML if portfolio == "BIG MeBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	1045.11744	3	348.372479	F(3, 1130)	=	4543.03
Residual	86.6516379	1,130	.076682865	Prob > F	=	0.0000
				R-squared	=	0.9234
				Adj R-squared	=	0.9232
Total	1131.76908	1,133	.998913571	Root MSE	=	.27692

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	.9050456	.0082036	110.32	0.000	.8889496	.9211417
SMB	.0306078	.0139002	2.20	0.028	.0033347	.0578809
HML	.3255254	.0086746	37.53	0.000	.3085053	.3425455
_cons	-.0101792	.0082366	-1.24	0.217	-.02634	.0059817

```
. regress vwreRF MktRF SMB HML if portfolio == "BIG HiBM"
```

Source	SS	df	MS	Number of obs	=	1,134
Model	1682.39257	3	560.797522	F(3, 1130)	=	3704.56
Residual	171.059753	1,130	.151380313	Prob > F	=	0.0000
				R-squared	=	0.9077
				Adj R-squared	=	0.9075
Total	1853.45232	1,133	1.63588025	Root MSE	=	.38908

vwreRF	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
MktRF	1.044091	.0115264	90.58	0.000	1.021475	1.066706
SMB	.0614881	.0195302	3.15	0.002	.0231686	.0998076
HML	.7505732	.0121881	61.58	0.000	.7266595	.774487
_cons	.0007219	.0115727	0.06	0.950	-.0219846	.0234283

Table E6 Time-series regression outputs for each portfolio (Kenneth R. French daily data). The vwreRF and MktRF are respectively portfolio value-weighted average return minus risk-free rate and market return minus risk-free rate

Breusch and Pagan Lagrangian multiplier test for random effects

```
vwret_risk_premium[eportfolio,t] = Xb + u[eportfolio] + e[eportfolio,t]
```

Estimated results:

	Var	SD = sqrt(Var)
vwret_r~m	.0002964	.0172157
e	.0000855	.0092472
u	0	0

Test: Var(u) = 0

```
chibar2(01) = 0.00
Prob > chibar2 = 1.0000
```

Table E7 Breusch and Pagan Lagrangian multiplier test for random effects (daily data)

Source	SS	df	MS	Number of obs	=	6,804
				F(3, 6800)	=	5593.88
Model	1.43486067	3	.47828689	Prob > F	=	0.0000
Residual	.581412607	6,800	.000085502	R-squared	=	0.7116
				Adj R-squared	=	0.7115
Total	2.01627328	6,803	.00029638	Root MSE	=	.00925

vwret_risk_premium	Coefficient	Std. err.	t	P> t	[95% conf. interval]
Market_RiskPremium	1.295578	.0117557	110.21	0.000	1.272534 1.318623
SMB	.5202356	.0207647	25.05	0.000	.4795304 .5609408
HML	.1317942	.0172608	7.64	0.000	.0979576 .1656308
_cons	.0000527	.0001122	0.47	0.639	-.0001673 .0002727

1

Coefficients: generalized least squares
Panels: homoskedastic
Correlation: no autocorrelation

Estimated covariances	=	1	Number of obs	=	6,804
Estimated autocorrelations	=	0	Number of groups	=	6
Estimated coefficients	=	4	Time periods	=	1134
			Wald chi2(3)	=	16791.50
Log likelihood	=	22213.98	Prob > chi2	=	0.0000

vwret_risk_premium	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Market_RiskPremium	1.295578	.0117522	110.24	0.000	1.272544	1.318612
SMB	.5202356	.0207585	25.06	0.000	.4795496	.5609216
HML	.1317942	.0172558	7.64	0.000	.0979735	.1656149
_cons	.0000527	.0001122	0.47	0.639	-.0001672	.0002726

Table E9 FGLS model (daily data)

Augmented Dickey-Fuller test for unit root

[illegible]

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-18.947	-2.580	-1.950	-1.620

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-18.807	-2.580	-1.950	-1.620

```
Variable: Market_RiskPrem      Number of obs = 648
                                Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.336	-2.580	-1.950	-1.620

```
Variable: small_low      Number of obs = 648
                        Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.480	-2.580	-1.950	-1.620

```
Variable: small_medium      Number of obs = 648
                             Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.624	-2.580	-1.950	-1.620

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.980	-2.580	-1.950	-1.620

```
Variable: big_low      Number of obs = 648
                        Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-18.049	-2.580	-1.950	-1.620

```
Variable: big_medium      Number of obs = 648
                          Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.562	-2.580	-1.950	-1.620

```
Variable: big_high      Number of obs = 648
                        Number of lags = 1
```

H_0 : Random walk without drift, $a = 0$, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-17.572	-2.580	-1.950	-1.620

Table E10 Augmented Dickey-Fuller test for stationary (unit root) test (daily data)

```
. vif
```

Variable	VIF	1/VIF
SMB	1.58	0.631129
HML	1.52	0.656820
Market_Risk	1.15	0.871111
Mean VIF	1.42	

Table E11 Variance Inflation Factor test for Multicollinearity (daily data)

Small-low portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 1130) = 2.04
Prob > F = 0.1070
```

Table E12 Breusch-Pagan test for Heteroskedasticity (small-low portfolio-daily data)

```
. estat dwatson

Number of gaps in sample = 246

Durbin-Watson d-statistic( 4, 1134) = 1.698367
```

Table E13 Durbin-Watson test for Autocorrelation (small-low portfolio-daily data)

```
. jb resid
Jarque-Bera normality test: 688.5 Chi(2) 3.e-150
Jarque-Bera test for Ho: normality:
```

Table E14 Jarque-Bera test for Normality (small-low portfolio-daily data)

Small-medium portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(3, 1130) = 2.31
Prob > F = 0.0750
```

Table E15 Breusch-Pagan test for Heteroskedasticity (small-medium portfolio-daily data)

```
. estat dwatson
```

```
Number of gaps in sample = 246
```

```
Durbin-Watson d-statistic( 4, 1134) = 1.526646
```

Table E16 Durbin-Watson test for Autocorrelation (small-medium portfolio-daily data)

```
. jb resid
```

```
Jarque-Bera normality test: 799.4 Chi(2) 3.e-174
```

```
Jarque-Bera test for Ho: normality:
```

Table E17 Jarque-Bera test for Normality (small-medium portfolio-daily data)

Small-high portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
```

```
Assumption: i.i.d. error terms
```

```
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 1130) = 4.15
```

```
Prob > F = 0.0061
```

Table E18 Breusch-Pagan test for Heteroskedasticity (small-high portfolio-daily data)

```
. estat dwatson
```

```
Number of gaps in sample = 246
```

```
Durbin-Watson d-statistic( 4, 1134) = 1.572365
```

Table E19 Durbin-Watson test for Autocorrelation (small-high portfolio-daily data)

```
. jb resid
```

```
Jarque-Bera normality test: 5282 Chi(2) 0
```

```
Jarque-Bera test for Ho: normality:
```

Table E20 Jarque-Bera test for Normality (small-high portfolio-daily data)

Big-low portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
```

```
Assumption: i.i.d. error terms
```

```
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 1130) = 1.00
```

```
Prob > F = 0.3940
```

Table E21 Breusch-Pagan test for Heteroskedasticity (big-low portfolio-daily data)

```
. estat dwatson
```

```
Number of gaps in sample = 246
```

```
Durbin-Watson d-statistic( 4, 1134) = 1.529287
```

Table E22 Durbin-Watson test for Autocorrelation (big-low portfolio-daily data)

```
. jb resid
```

```
Jarque-Bera normality test: 615.9 Chi(2) 2.e-134
```

```
Jarque-Bera test for Ho: normality:
```

Table E23 Jarque-Bera test for Normality (big-low portfolio-daily data)

Big-medium portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
```

```
Assumption: i.i.d. error terms
```

```
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 1130) = 2.40
```

```
Prob > F = 0.0665
```

Table E24 Breusch-Pagan test for Heteroskedasticity (big-medium portfolio-daily data)

```
. estat dwatson
```

```
Number of gaps in sample = 246
```

```
Durbin-Watson d-statistic( 4, 1134) = 1.47074
```

Table E25 Durbin-Watson test for Autocorrelation (big-medium portfolio-daily data)

```
. jb resid
```

```
Jarque-Bera normality test: 296.9 Chi(2) 3.4e-65
```

```
Jarque-Bera test for Ho: normality:
```

Table E26 Jarque-Bera test for Normality (big-medium portfolio-daily data)

Big-high portfolio:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
```

```
Assumption: i.i.d. error terms
```

```
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 1130) = 0.27
```

```
Prob > F = 0.8455
```

Table E27 Breusch-Pagan test for Heteroskedasticity (big-high portfolio-daily data)

```
. estat dwatson
```

```
Number of gaps in sample = 246
```

```
Durbin-Watson d-statistic( 4, 1134) = 1.669378
```

Table E28 Durbin-Watson test for Autocorrelation (big-high portfolio-daily data)

```
. jb resid
```

```
Jarque-Bera normality test: 191.7 Chi(2) 2.4e-42
```

```
Jarque-Bera test for Ho: normality:
```

Table E29 Jarque-Bera test for Normality (big-high portfolio-daily data)

Diagnostic tests or panel regression (POLS)

Levin-Lin-Chu unit-root test for Market_RiskPremium

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common	Asymptotics: N/T ->	0
Panel means: Included		
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-58.4641	
Adjusted t*	-62.5999	0.0000

Levin-Lin-Chu unit-root test for SMB

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common	Asymptotics: N/T ->	0
Panel means: Included		
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-57.6854	
Adjusted t*	-61.7269	0.0000

Levin-Lin-Chu unit-root test for HML

H0: Panels contain unit roots Number of panels = 6
 Ha: Panels are stationary Number of periods = 1134

 AR parameter: Common Asymptotics: N/T -> 0
 Panel means: Included
 Time trend: Not included

 ADF regressions: 1 lag
 LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-57.1884	
Adjusted t*	-61.5941	0.0000

Levin-Lin-Chu unit-root test for vwret_risk_premium

H0: Panels contain unit roots Number of panels = 6
 Ha: Panels are stationary Number of periods = 1134

 AR parameter: Common Asymptotics: N/T -> 0
 Panel means: Included
 Time trend: Not included

 ADF regressions: 1 lag
 LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-55.5635	
Adjusted t*	-57.3292	0.0000

Table E30 Levin-Lin-Chu test for stationary (unit root) test (daily data)

. vif

Variable	VIF	1/VIF
SMB	1.58	0.631129
HML	1.52	0.656820
Market_Ris~m	1.15	0.871111
Mean VIF	1.42	

Table E31 Variance Inflation Factor test for Multicollinearity (daily data)

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
 Assumption: i.i.d. error terms
 Variables: All independent variables

H0: Constant variance

$F(3, 6800) = 6.09$
 Prob > F = 0.0004

Table E32 Breusch-Pagan test for Heteroskedasticity (daily data)

. jb resid
 Jarque-Bera normality test: 4.9e+05 Chi(2) 0
 Jarque-Bera test for Ho: normality:

Table E33 Jarque-Bera test for Normality (daily data)

Wooldridge test for autocorrelation in panel data
 H0: no first order autocorrelation
 $F(1, 5) = 5.139$
 Prob > F = 0.0727

Table E34 Wooldridge test for panel data Autocorrelation (daily data)

F Statistical Outputs (daily data with Bitcoin from July 2018 to December 2022)

Breusch and Pagan Lagrangian multiplier test for random effects

`Lvwret_risk_premium[eportfolio,t] = Xb + u[eportfolio] + e[eportfolio,t]`

Estimated results:

	Var	SD = sqrt(Var)
Lvwret_~m	1.336277	1.155974
e	.7969871	.8927414
u	0	0

Test: Var(u) = 0

`chibar2(01) = 0.00`

`Prob > chibar2 = 1.0000`

Table F1 Breusch and Pagan Lagrangian multiplier test for random effects (daily data with Bitcoin). Lvwret_risk_premium stands for log(vwret_risk_premium)

`. grstest2 Lbig_high-Lsmall_medium, flist(LMarket_RiskPremium LSMB LHML Lbitcoin_return) alphas`

R[2,7]

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	-1.35535	93.222657	0	.34868668	.78395409	1.5270544	0
J1	-1.35535	16.176873	5.089e-10	.34868668	.78395409	1.5270544	8.9646267

Table F2 GRS-test outputs (daily data with Bitcoin). LMarket_RiskPremium, LSMB, LHML, and Lbitcoin_return stand for log (market risk premium), log (SMB), log (HML), and log (Bitcoin return)

`. regress Lvwret_risk_premium LMarket_RiskPremium LSMB LHML Lbitcoin_return`

Source	SS	df	MS	Number of obs	=	373
Model	100.324115	4	25.0810287	F(4, 368)	=	23.26
Residual	396.770856	368	1.07818167	Prob > F	=	0.0000
				R-squared	=	0.2018
				Adj R-squared	=	0.1931
Total	497.094971	372	1.3362768	Root MSE	=	1.0384

Lvwret_risk_premium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
LMarket_RiskPremium	.368293	.049245	7.48	0.000	.2714562	.4651299
LSMB	.2120203	.0532564	3.98	0.000	.1072952	.3167454
LHML	-.0462344	.0575878	-0.80	0.423	-.159477	.0670081
Lbitcoin_return	.1344388	.0440313	3.05	0.002	.0478543	.2210233
_cons	-1.30372	.394977	-3.30	0.001	-2.080415	-.5270251

Table F3 Panel regression output – POLS model (daily data with Bitcoin). LMarket_RiskPremium, LSMB, LHML, and Lbitcoin_return stand for log (market risk premium), log (SMB), log (HML), and log (Bitcoin return)

Cross-sectional time-series FGLS regression

Coefficients: generalized least squares

Panels: homoskedastic

Correlation: no autocorrelation

Estimated covariances	=	1	Number of obs	=	373
Estimated autocorrelations	=	0	Number of groups	=	6
Estimated coefficients	=	5	Obs per group:		
			min	=	44
			avg	=	62.16667
			max	=	78
			Wald chi2(4)	=	94.31
Log likelihood	=	-540.7861	Prob > chi2	=	0.0000

Lvwret_risk_premium	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
LMarket_RiskPremium	.368293	.0489138	7.53	0.000	.2724237	.4641623
LSMB	.2120203	.0528983	4.01	0.000	.1083416	.315699
LHML	-.0462344	.0572006	-0.81	0.419	-.1583455	.0658766
Lbitcoin_return	.1344388	.0437352	3.07	0.002	.0487195	.2201581
_cons	-1.30372	.3923207	-3.32	0.001	-2.072655	-.5347857

Table F4 FGLS model for correcting the Heteroskedasticity and Autocorrelation (daily data with Bitcoin). LMarket_RiskPremium, LSMB, LHML, and Lbitcoin_retun stand for log (market risk premium), log (SMB), log (HML), and log (Bitcoin return)

Diagnostic tests for panel regression (Random Effect Model):

Levin-Lin-Chu unit-root test for LMarket_RiskPremium

H0: Panels contain unit roots	Number of panels	=	6
Ha: Panels are stationary	Number of periods	=	606

AR parameter: Common	Asymptotics: N/T -> 0
Panel means: Included	
Time trend: Not included	

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-17.8074	
Adjusted t*	-8.2515	0.0000

```
. xtunitroot llc LSMB
```

Levin-Lin-Chu unit-root test for LSMB

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	565

AR parameter: Common	Asymptotics: N/T -> 0
Panel means: Included	
Time trend: Not included	

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-23.8914	
Adjusted t*	-14.0272	0.0000

Levin-Lin-Chu unit-root test for LHML

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	539

AR parameter: Common	Asymptotics: N/T -> 0
Panel means: Included	
Time trend: Not included	

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-13.7715	
Adjusted t*	-0.7597	0.2237

Levin-Lin-Chu unit-root test for Lbitcoin_return

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	570

AR parameter: Common	Asymptotics: N/T -> 0
Panel means: Included	
Time trend: Not included	

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-18.0043	
Adjusted t*	-8.1871	0.0000

Table E5 Levin-Lin-Chu test for stationary (unit root) test (daily data with Bitcoin)

```
. vif
```

Variable	VIF	1/VIF
LSMB	1.37	0.727758
LHML	1.35	0.738170
LMarket_Return	1.04	0.959093
Lbitcoin_return	1.01	0.986368
Mean VIF	1.20	

Table F6 Variance Inflation Factor test for Multicollinearity (daily data with Bitcoin)

```
. hettest, rhs fstat
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(4, 368) = 4.79
Prob > F = 0.0009

=====

*** Panel Groupwise Heteroscedasticity Tests**

=====

Ho: Panel Homoscedasticity - Ha: Panel Groupwise Heteroscedasticity

- Lagrange Multiplier LM Test	=9143.7779	P-Value > Chi2(5)	0.0000
- Likelihood Ratio LR Test	= 426.0342	P-Value > Chi2(5)	0.0000
- Wald Test	= 2.27e+04	P-Value > Chi2(6)	0.0000

Table F7 Wald and Breusch-Pagan test for Heteroskedasticity (daily data with Bitcoin). Both tests show Heteroskedasticity

```
. jb resid
```

Jarque-Bera normality test: 144.6 Chi(2) 3.9e-32
Jarque-Bera test for Ho: normality:

Table F8 Jarque-Bera test for Normality (daily data with Bitcoin)

Wooldridge test for autocorrelation in panel data

H0: no first order autocorrelation

F(1, 1) = 26.600
Prob > F = 0.1219

Table F9 Wooldridge test for panel data Autocorrelation (daily data with Bitcoin)

```
. regress Lvwret_risk_premium LMarket_RiskPremium LSMB Lbitcoin_return
```

Source	SS	df	MS	Number of obs	=	703
Model	168.825957	3	56.2753191	F(3, 699)	=	56.51
Residual	696.11082	699	.995866695	Prob > F	=	0.0000
				R-squared	=	0.1952
				Adj R-squared	=	0.1917
Total	864.936777	702	1.23210367	Root MSE	=	.99793

Lvwret_risk_premium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
LMarket_RiskPremium	.4102079	.0348896	11.76	0.000	.3417069	.4787088
LSMB	.1060771	.0311681	3.40	0.001	.0448827	.1672714
Lbitcoin_return	.1092264	.0305499	3.58	0.000	.0492458	.169207
_cons	-1.489979	.2782129	-5.36	0.000	-2.036212	-.9437456

Table F10 The POLS output (daily data with Bitcoin) after removing the LHML. LMarket_RiskPremium, LSMB, and Lbitcoin_return stand for log (market risk premium), log (SMB), and log (Bitcoin return)

Cross-sectional time-series FGLS regression

Coefficients: generalized least squares

Panels: homoskedastic

Correlation: no autocorrelation

Estimated covariances	=	1	Number of obs	=	703
Estimated autocorrelations	=	0	Number of groups	=	6
Estimated coefficients	=	4	Obs per group:		
			min	=	97
			avg	=	117.1667
			max	=	133
			Wald chi2(3)	=	170.50
Log likelihood	=	-994.0522	Prob > chi2	=	0.0000

Lvwret_risk_premium	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
LMarket_RiskPremium	.4102079	.0347902	11.79	0.000	.3420203	.4783954
LSMB	.1060771	.0310793	3.41	0.001	.0451627	.1669914
Lbitcoin_return	.1092264	.0304629	3.59	0.000	.0495203	.1689325
_cons	-1.489979	.2774203	-5.37	0.000	-2.033712	-.9462449

Table F11 The FGLS output (daily data with Bitcoin) after removing the LHML. LMarket_RiskPremium, LSMB, and Lbitcoin_return stand for log (market risk premium), log (SMB), and log (Bitcoin return)

Breusch and Pagan Lagrangian multiplier test for random effects

`vwret_risk_premium[eportfolio,t] = Xb + u[eportfolio] + e[eportfolio,t]`

Estimated results:

	Var	SD = sqrt(Var)
vwret_risk_premium	.0007646	.027651
e	.0005554	.0235669
u	0	0

Test: Var(u) = 0

`chibar2(01) = 0.00`
 Prob > chibar2 = 1.0000

Table F12 Breusch and Pagan Lagrangian multiplier test for random effects (daily data without Bitcoin for 29 firms). The `vwret_risk_premium` stands for value-weighted average return minus risk-free rate

`. grstest2 big_high-small_medium, flist(Market_RiskPremium SMB HML) alphas`

R[2,7]

	Mean alpha	Test statistic	P-value	Mean adj R2	Mean SE	Mean abs alpha	SR
J0	.00011444	3.2912842	.77149895	.71437328	.00025076	.00011444	0
J1	.00011444	.54419381	.77476699	.71437328	.00025076	.00011444	.05391659

Table F13 GRS-test outputs (daily data without Bitcoin for 29 firms)

`. regress vwret_risk_premium Market_RiskPremium SMB HML`

Source	SS	df	MS	Number of obs	=	6,804
Model	1.4262256	3	.475408535	F(3, 6800)	=	856.32
Residual	3.77520707	6,800	.000555178	Prob > F	=	0.0000
				R-squared	=	0.2742
				Adj R-squared	=	0.2739
Total	5.20143268	6,803	.000764579	Root MSE	=	.02356

vwret_risk_premium	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Market_RiskPremium	1.405855	.0282307	49.80	0.000	1.350514	1.461196
SMB	.2471433	.0346331	7.14	0.000	.1792516	.315035
HML	.1320404	.0244052	5.41	0.000	.0841986	.1798822
_cons	.0002147	.0002859	0.75	0.453	-.0003457	.0007751

Table F14 Panel regression output – POLS model (daily data without Bitcoin for 29 firms)

Cross-sectional time-series FGLS regression

Coefficients: generalized least squares

Panels: homoskedastic

Correlation: no autocorrelation

Estimated covariances	=	1	Number of obs	=	6,804
Estimated autocorrelations	=	0	Number of groups	=	6
Estimated coefficients	=	4	Time periods	=	1134
			Wald chi2(3)	=	2570.47
Log likelihood	=	15849.69	Prob > chi2	=	0.0000

vwret_risk_premium	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Market_RiskPremium	1.405855	.0282224	49.81	0.000	1.35054	1.46117
SMB	.2471433	.0346229	7.14	0.000	.1792837	.3150029
HML	.1320404	.024398	5.41	0.000	.0842212	.1798596
_cons	.0002147	.0002858	0.75	0.453	-.0003455	.0007748

Table F15 FGLS model for correcting the Heteroskedasticity and Autocorrelation (daily data without Bitcoin for 29 firms)

Levin-Lin-Chu unit-root test for Market_RiskPremium

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common	Asymptotics: N/T ->	0
Panel means: Included		
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-58.4641	
Adjusted t*	-62.5999	0.0000

Levin-Lin-Chu unit-root test for SMB

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common	Asymptotics: N/T ->	0
Panel means: Included		
Time trend: Not included		

ADF regressions: 1 lag

LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)

	Statistic	p-value
Unadjusted t	-57.5066	
Adjusted t*	-61.5407	0.0000

Levin-Lin-Chu unit-root test for HML

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common		
Asymptotics: N/T -> 0		
Panel means: Included		
Time trend: Not included		
ADF regressions: 1 lag		
LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)		

	Statistic	p-value
Unadjusted t	-58.6904	
Adjusted t*	-63.1196	0.0000

Levin-Lin-Chu unit-root test for bitcoin_return

H0: Panels contain unit roots	Number of panels =	6
Ha: Panels are stationary	Number of periods =	1134
AR parameter: Common		
Asymptotics: N/T -> 0		
Panel means: Included		
Time trend: Not included		
ADF regressions: 1 lag		
LR variance: Bartlett kernel, 33.00 lags average (chosen by LLC)		

	Statistic	p-value
Unadjusted t	-57.4550	
Adjusted t*	-61.1996	0.0000

Table F16 Levin-Lin-Chu test for stationary (unit root) test (daily data for 29 firms)

Variable	VIF	1/VIF
HML	1.40	0.716700
SMB	1.37	0.728565
Market_Ris~m	1.02	0.980806
Mean VIF	1.26	

Table F17 Variance Inflation Factor test for Multicollinearity (daily data without Bitcoin for 29 firms)

Wooldridge test for autocorrelation in panel data
H0: no first order autocorrelation
F(1, 5) = 10.709
Prob > F = 0.0221

Table F18 Wooldridge test for panel data Autocorrelation (daily data without Bitcoin for 29 firms)

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity  
Assumption: i.i.d. error terms  
Variables: All independent variables
```

```
H0: Constant variance
```

```
F(3, 6800) = 2.89  
Prob > F = 0.0342
```

Table F19 Breusch-Pagan test for Heteroskedasticity (daily data without Bitcoin for 29 firms)

```
. jb resid  
Jarque-Bera normality test: 2.2e+07 Chi(2) 0  
Jarque-Bera test for Ho: normality:
```

Table F20 Jarque-Bera test for Normality (daily data without Bitcoin for 29 firms)

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